

A Wearable-Based Aging Clock Associates with Disease and Behavior

Corresponding Author: Dr Andrew Miller

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Summary: The article titled "A Wearable-Based Biomarker of Aging and Its Association With Disease and Behavior" implements a novel self-supervised learning (SSL) approach to predict chronological age utilizing continuously monitored PPG data collected from consumer wearables in a cohort of AHMS. Additionally, it assesses the associations between the estimated aging measure, denoted as PpgAge, and disease diagnoses, smoking, and sleep stages. Furthermore, it demonstrates a substantial increase in PpgAge during pregnancy in women from a longitudinal analysis. Overall, the paper provides insights into utilizing data from consumer wearables combined with DL approach to predict chronological age and show its associations with behavioral and physiological measures.

Strengths:

- 1 – SSL has gained increasing attention and shown potentials in recent studies, some of particularly those using longitudinal, continuous data from wearable devices. The article is interesting and innovative by applying this DL approach to a new modality that has not been previously explored in the context of biological age estimation.
- 2 – This investigation aids readers in comprehending and identifying the possibilities of consumer devices in studying longevity or aging. Given the growing adoption of consumer wearables, such investigation holds potential for health monitoring and clinical usage from various angles, if scientific rigor and validation is warranted.

Weaknesses:

- 1 – It is interesting to know that PpgAge has a high accuracy of predicting chronological age. However, the paper lacks evaluation of PpgAge in predicting other age-related outcomes such as mortality, incident diseases, or functional outcomes. The authors should demonstrate the additional PpgAge's predictive ability with these outcome(s).
- 2 – In the introduction, better explanation and motivation of using PPG and SSL in age prediction are needed. Also, please consider that different wearable modalities may reflect different aspects of aging, e.g., PPG may be more reflective to vascular aging.
- 3 – In methods, detailed information on the detailed breakdowns of types of consumer wearables (i.e., Apple Watch) and software versions used to collect the PPG data. Are all participants used the same smartwatch model during the study follow-up? This information is crucial, particularly with consumer wearables where hardware/software may vary largely.
- 4 – Is there information on the comparison or ascertainment of the Apple Watch's PPG to the ground truth? Such validation is highly important when studying with wearables.
- 5 – Data processing lacks critical details on wear time, data quality, handling missing data, etc. Because SLL may mask or crop to remove parts, which can be problematic and may undermine abnormalities in case of insufficient monitoring period. Please provide a through explanation of the data preprocessing workflow.
- 6 – In the discussion, it is unclear how PpgAge could be interpreted or used in practical settings. It lacks discussion the explainability of model predictions, leaving questions about the practical implications of PpgAge and PpgAge gap. For example, if an individual is estimated with PpgAge gap of +2 years, thus "seemingly old", what drives or constitutes this age gap value? What is the practical implication for this specific individual?

Specific comments

- 1 – Title: I think the study's central focus of the paper is chronological age prediction applying SSL to PPG data. 'wearable-based biomarker of aging' may be interpreted as an overstatement given the current results. Please consider revising the title to more accurately reflect the study.

2 – Results

- (i) For results presenting using quartile groups, please provide the exact cut-off values.
- (ii) What is the median (IQR) follow-up time? Please also include information on withdrawal or dropouts during follow up.
- (iii) What are the cohort characteristics? Provide a table including baseline characteristics.
- (iv) PpgAge gap seems not normally distributed. It would be informative to investigate why and include discussion.
- (v) Please consider whether employment, shift work, and chronotype impact the associations in sleep analysis. For example, present results on the employment type, shift work status, or chronotype. And/or consider excluding shift workers from the analysis.
- (vi) In the association of PpgAge and pregnancy,
 - a) Is the PpgAge increase at the same rate across different age groups of mothers?
 - b) Please consider sensitivity analyses stratifying by/controlling for healthy vs. pre-condition status, BMI, comorbidities, socioeconomic status, and demographic factors.
 - c) In pg. 6, "we observe a slightly negative median age gap slope for a year post parturition", does this imply that childbirth is associated with a permanent increase in biological age for women, even if they are otherwise healthy and disease-free?
- (vii) Figure 4(a)-(b) captions do not seem to match the text. (pg. 4)
- (viii) It is unclear how "exercise minutes" are quantified (pg. 5). Is this using MET or MVPA? Please provide detailed explanation.

3 – Methods

- (i) Please include more detailed information on the model evaluation e.g., data split for training and evaluation and internal validation.
- (ii) Please clarify what packages, at what stage, were used in R and python.
- (iii) External validation in another dataset is strongly encouraged.

(Remarks on code availability)

N/A

Reviewer #3

(Remarks to the Author)

The authors took advantage of data collected with Apple watches from 200 thousands participants to estimate a measure of biological age (also called "aging clocks") through the waves recorded by photoplethysmography (PPG), elaborated using a machine learning approach and setting chronological age as the "output" variable to build the clock. The idea per se is really interesting, and, as reported by the authors, in line with the ongoing research efforts aimed at finding a measure of biological age able to implement risk stratification for common age-related diseases (and/or mortality) on top of canonical risk factors without expensive or complex analyses. Thus, the topic is of high interest and the general approach is meritorious. The manuscript is well-written. However, I have some doubts relative to some passages of the paper and, more broadly, on the basic idea itself. In detail:

1- The authors quickly define the input measure (with a couple of lines in the introduction). Researchers in the field of aging or CVD might not know what is a photoplethysmography and how it works and especially what it's "sensing". It is stated that it detects changes in blood volumes but I don't think this is a precise definition. I guess it senses blood volumes in a very specific area (the microvascular bed) and not at the organismal level. Thus, it basically senses the movement of blood through small arteries in the arm. This is likely the result of a combination of heart rate and blood pressure (which reflects arterial elasticity and overall blood volume, among others). Thus, such measure is very intrinsically linked to the health of the cardiovascular system. As a result, it is not surprising that it performs very well in the Cross-sectional detection of CVD, but fails to detect other age-related diseases (e.g. cancer, as reported by the authors). This implies that such an approach might be better suited to detect cardiovascular aging, rather than overall organism-level, biological aging. Indeed, even though the geroscience hypothesis suggests that few, common aging pathways underlie the development of all age-related diseases, recent manuscripts in the field suggest that different organs might age at a different rate within the same person and among individuals (DOI: 10.1016/j.arr.2024.102253; <https://doi.org/10.1038/s41576-022-00511-7>). Thus, I suggest to: 1- better explain in the introduction what is measured by PPG and what determines such waves; 2- Tone down some of the statements relative to the ability of this tool to detect biological aging; 3- Discuss (in the discussion section) the possibility that this tool might work well with cardiovascular conditions but not with other age-related diseases and, more broadly, that such an approach is intrinsically linked to cardiovascular health.

2- For the same reason, the authors tried to compare their approach with an alternative one taking advantage of heart rate and heart rate variability, which are measured by the watch through the same technology. Their approach outperformed the other in detecting chronological age but this aspect is not discussed. Why the chosen approach is superior? Which biological aspect is missed when using only heart rate? Arterial elasticity? Blood pressure? Please discuss briefly in the discussion section.

3- The authors used chronological age as the variable to build the clock. This approach has been often used but available data (from epigenetic, metabolomic, or proteomic-based clocks) suggest that this approach is very precise in estimating chronological age but has a poor performance in longitudinally identifying the incidence of age-related disease or mortality (DOI: 10.1016/j.arr.2024.102253; <https://doi.org/10.1038/s41576-022-00511-7>), which should be the final goal of metrics of biological age. The results presented indeed sustain a cross-sectional association with some diseases, but only a

preliminary analysis is shown relatively to the longitudinal association with CVD (and pregnancy, which is not a disease). The data relatively to the longitudinal incidence of CVD should be better presented and explained. Which is the delta of years associated with an increased risk of CVD? Which are the relative hazard ratios? Does it predict CVD also when adjusting for cardiovascular risk factors (e.g. LDL cholesterol, blood pressure, HbA1c in case of diabetes) or it improves risk stratification of established models when estimated age is used instead of calendar age (or the delta is added as a further covariate)? Please perform and present additional analysis on this aspect.

In addition, it is unclear why a measure that should predict long-term outcomes (i.e. biological age and age related diseases) should change in the close proximity of the event (the authors evaluated the 180 days before a CVD event). This is not in line with the idea of biological age and with the mentioned geroscience hypothesis. Indeed, biological age should have a long-term performance, in order to allow the application of preventive strategies (e.g. statins, anti-platelet drugs, exercise, dieting) in patients at high risk that were classified not at high risk through conventional risk factors (e.g. blood pressure and LDL). If the measure changes in the close proximity of the event, it gives the impression of something closely linked to the condition but not really measuring the biological age of the individual. Also, what if a larger period of time is considered to estimate the measure of age or the delta (e.g. 2 years of measures instead of 180 days)? The performance in predicting CVD is the same? Please present the results with more granularity and discuss accordingly.

4- Relatively to the results in type 2 diabetes (T2D), it is known that this condition increases the glycation of protein and promotes the accumulation of advanced glycation end products. In turn, AGE accumulation is associated with an alteration of skin autofluorescence (SAF) which can be measured by specific instruments and correlates with multiple cardiovascular conditions in T2D. In there the possibility that the optical reader of PPG is affected by this phenomenon? This would not represent an issue per se but, again, it would link the approach to a specific condition. If my question is appropriate, please discuss. If not (I'm not an expert of PPG), please ignore my suggestion.

(Remarks on code availability)

The statement in the paper reports:

This code may be available upon request from the corresponding author (A.C.M.). Requests for code will be evaluated and responded to in a manner consistent with policies intended to protect participant confidentiality and language in the study protocol and informed consent form.

Thus codes are not available, as well as individual level patients data.

Reviewer #4

(Remarks to the Author)

In summary, may I thank the authors and their institutions for this manuscript on photoplethysmography-based assessment of biological aging, which I have no doubt will make a significant contribution to the field. In this report I highlight what I perceive to be the key strengths and weaknesses of the study, and provide some recommendations of areas for improvement. Please note that I have some experience in photoplethysmogram (PPG) signal processing, but less experience in statistical analyses, so my comments are predominantly focused on the PPG-related aspects of the manuscript.

KEY RESULTS

The authors present a large-scale study of 'Ppg-Age', a parameter derived from wrist photoplethysmography signals which is designed to predict chronological age. Ppg-Age is found to correlate well with chronological age. In addition, higher a 'Ppg-Age gap' (i.e. Ppg-Age minus chronological age) is found to be associated with: (i) higher rates of historical (rather than future) heart disease and diabetes diagnoses; (ii) lifestyle behaviours (namely smoking status, "exercise minutes", and sleep characteristics. Finally, 'Ppg-Age' appears to increase more rapidly in the latter stages of pregnancy than either early in pregnancy or soon after giving birth.

VALIDITY

There are limitations to the quality of the study data, including:

- PPG signals were acquired using reflectance photoplethysmography and green light using a device with potentially variable contact pressure and shielding from ambient light. I don't consider this to be a substantial limitation, as "dark subtraction" was used to attempt to mitigate against the effects of ambient light, and the performance reported will reflect (no pun intended) the limitations of the pulse wave sensor.

- Several outcomes were obtained from self-reported data, which may introduce inaccuracies. This was noted by the authors, and would likely reduce the ability of Ppg-Age to predict chronological age, and reduce the strengths of the associations with outcomes. Therefore, I do not see this as a substantial limitation.

- Little physiological interpretation of 'Ppg-Age' is provided (besides mentioning that "The full PPG waveform reflects changes in blood volume in the microvascular bed of tissue"). This is acknowledged by the authors, and again does not in my view represent a substantial limitation.

- Comment 1: There is little analysis of how Ppg-Age changes within an individual over time (i.e. whether it increases by an average of 1 year per chronological year), and I thought this could have greatly enhanced the study. Furthermore, I envisage this would be feasible given that some participants contributed data for four years.

SIGNIFICANCE

In my view this study is highly significant in several regards. First, it continues the narrative that large-scale studies of physiology and clinical outcomes can be conducted using consumer wearables, leading to valuable insights. Second, it demonstrates that it is possible to predict chronological age relatively accurately from wrist PPG in a large cohort. Third, it identifies potential bases for using PPG-derived parameters of biological age (namely associations with disease diagnosis, associations with lifestyle behaviours, and changes in pregnancy).

DATA AND METHODOLOGY

I have already commented on the limitations of the study data. I think the study appears to be well-conducted, the manuscript is well-presented, and unfortunately I do not feel qualified to judge some of the statistical methodology. However, I would highlight the following:

- Comment 2: I think the use of 'mean absolute error' to quantify agreement between Ppg-Age and chronological age is useful, but that it could be supplemented by additional statistics to give greater understanding of the level of agreement. Whilst I acknowledge r , r^2 , and $rmse$ are already used, I think these also do not provide a complete view. I would recommend reporting the bias and limits of agreement (as proposed by Bland and Altman), to help understand the accuracy and precision of Ppg-Age.
- Comment 3: Training and Testing data: I was confused about what data were used in pre-training and training, and what data were used to obtain the results. For instance, the manuscript states that "We pre-trained our PPG foundation model using 19,993,427 PPG segments from 172,318 participants". Please clarify whether some/all of these 172,318 participants are also used for training and/or to obtain the results. If so, please comment on the appropriateness of this (I am unfamiliar with whether such information leakage would be a limitation or not). Second, I get the impression that the model was trained on the "healthy" cohort, which in principle I see as acceptable as results are reported for the separate "general" cohort. However, in practice, the headline result appears to be "PpgAge predicts chronological age with high accuracy in both the "healthy" and "general" group", without an explicit acknowledgment at that point in the manuscript that the "healthy" cohort were used for training. Furthermore, the next sentence states "Prediction accuracy is highest in the healthy cohort" (in comparison to the general cohort). For this to be a fair comparison, presumably ideally it would be a comparison between healthy and general data which were not used for training (whereas in reality my understanding is that the healthy data were used in training, but the general data were not). Therefore: (i) if I have misunderstood, then please correct me and clarify the manuscript; or (ii) if I have understood that "healthy" cohort data were used for training, then please consider this when interpreting the results.
- Comment 4: I thought a strength of the study was the idea that one could train a model using chronological age as a reference. Much previous work has focused on using reference measurements which are reflective of the underlying ageing process (e.g. carotid-femoral pulse wave velocity), presumably based on the assumption that this would be more informative. However, the authors have developed a helpful model using chronological age alone for training. Perhaps you might consider whether to mention this aspect.
- Comment 5: Could you provide details of the equipment and algorithm used in this study, so that Apple Watch users can determine whether or not their watch measures PpgAge using the same approach reported in this study (e.g. same PPG sensor, and same algorithm)?
- Comment 6: Referring to Fig. 1(b), why are there: (i) fewer females in the 'All subjects' group (approx. 36% female); and (ii) substantially fewer females in the 'Healthy subjects' group (approx. 20%). Here, I think it would be helpful to provide insights into why: (i) more males than females enrolled in the study; and (ii) why the criteria for inclusion in the 'healthy subjects' group heightened this disparity.
- Comment 7: I am sceptical of the comparison between Ppg-Age and the comparator "heart rate and heart rate variability" model (see Results section titled 'PpgAge predictions use morphological features beyond heart rate and heart rate variability'). This is because the heart rate variability (HRV) model only included two HRV metrics, whereas many more metrics have been proposed in the literature to quantify HRV. Because of this, could it be that the conclusion that "HR/HRV-based models predict age less accurately than PpgAge across all subpopulations" is partly because the model only incorporates a small subset of the potential HRV parameters, and perhaps the difference in performance between Ppg-Age and HRV would be less if the HRV model included more parameters?
- Comment 8: In the analysis of associations between behaviours and Ppg-Age, it appears that some investigations (e.g. sleep, exercise behaviour, disease) involve matching participants (e.g. by age and sex) whereas the investigation into smoking does not. Is this the case? If so, why not do matching for smoking status? If not, please clarify the methodology in the manuscript. Also, is it sufficient to match by age and sex, or would it be helpful to also match by blood pressure given its impact on Ppg-Age?
- Comment 9: Please state the wavelength of green light used in the photoplethysmography sensor.

SUGGESTED IMPROVEMENTS

- Comment 10: I would suggest further discussing the potential physiological basis for Ppg-Age, including the authors'

perspectives on: (i) whether there is need to understand the physiology in order to use the parameter (notwithstanding their current remarks on "physiological mechanisms" and "longevity interventions"); and if so, (ii) potential directions for future research to further understand the physiological basis for PPG-derived markers of ageing (particularly reflectance PPG at the wrist). I would also suggest discussing the potential influence of blood pressure (see Fig. S2), and the (possible) benefit to understanding whether Ppg-Age provides additional information beyond that of blood pressure alone.

- Comment 11: Might the authors consider providing further suggestions as to the potential applications of Ppg-Age? Currently the Discussion suggests "effective longevity interventions" and "potential for clinical translation". It might be helpful to the reader to explore this a bit further, and in particular, consider whether any Ppg-Age functionality would need to be certified for medical use to be used clinically, and whether there is an expectation that consumers could respond to the Ppg-Age reported from their watch without clinical input.

CLARITY AND CONTEXT

The study is presented clearly, both in the text and figures, with a wealth of results presented in both the main text and supplementary material.

- Comment 12: The early stages of the manuscript (Abstract and Introduction) could give the impression that the assessment of biological age from the PPG is a novel idea. e.g. (i) "Existing measures of biological age are typically based on blood, saliva, vital signs, or clinical imaging. Wearables, however, can make convenient, frequent, and inexpensive measurements throughout daily life, scaled to an entire population."; (ii) the paragraph starting "Efforts to quantify biological aging" does not mention PPG-based approaches. I would recommend avoiding any possible impression that this idea is novel, because it is present in both the academic literature and wearable devices (e.g. Oura Ring's "cardiovascular age").

- Comment 13: I thought one key aspect of the context of the work was missing: there is minimal discussion of the prior art of assessing biological age from the PPG. This is a wide area of research (see our review - <https://doi.org/10.1152/ajpheart.00392.2021> for discussion of how wide this area of research is). A relatively novel aspect of the current study in comparison to previous work is that it uses widely monitored wrist-PPG signals, as opposed to finger-PPG signals (which I think more of the previous research has focused on). I would recommend discussing the prior art of PPG-based assessment of biological age in more detail.

- Comment 14: I saw no acknowledgments or details of funding provided in the manuscript. Whilst perhaps these are provided in the submission system rather than the manuscript itself, I would like the reader to be aware of who funded the study, and any conflicts of interest.

- Comment 15: Fig 5: These images appeared highly pixelated on the PDF I was given. Please check that these and the other images in the manuscript are clear, and provided in a suitable format.

REFERENCES

- Comment 16: It's unfortunate that Ref [App21], which is referred to in the context of the methodology used to estimate Vo2 max, refers to a webpage without a DOI. I wondered whether in light of this it might be helpful to provide more detail of the methodology used to estimate Vo2 max in the manuscript.

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Thank you for thoroughly addressing my comments. The additional information and analyses have clarified many initial questions. I have a few remaining comments and suggestions, mainly regarding the analyses of disease incidence, a key analysis to demonstrate the predictive accuracy of PpgAge.

1. Key details are missing in the analysis results:

- Please report the total number of events (N events) for each disease.
- When does person-time start and end in the analysis?
- Over what time frame are these events assessed? Is this an "ever" diagnosis or within a specific follow-up period?

2. In line 625, the authors state that disease diagnoses come from self-reports from repeated surveys and EHR.

- What questions (and responses) and medical codes are used for disease diagnoses from self-reports and electronic records/Medicare, respectively?
- Has any sensitivity analysis been conducted to confirm the reliability of self-reported diagnoses? A study reports that ~10% of AHMS participants have shared clinical health records (<https://doi.org/10.1038/s41746-024-01187-5>). If this holds true, the vast majority of disease diagnoses rely on self-reports from follow-up surveys, and the sensitivity of self-reported

incident diseases is shown to be substantially low, raising questions about the validity and accuracy of reporting. It may also be influenced by factors such as age or education level.

c) Please consider medication use in the identification of incident disease.

3. In Fig 4, the Cox PH models use a dichotomised group of >6 years vs. ≤6 years for PpgAge gap. It is unclear for choosing this 6-year cutoff in this analysis, especially since a continuous age gap may be more relevant for application and provide interpretable insights.

4. In Figure S1, PpgAge gap is relatively symmetric around 0 years, and the BA plots show $|1.96SD|$ of ~6 years in the healthy test set and ~8 years in the general population. This suggests that the individuals with PpgAge gap > 6 years are likely outliers, very fast-aging looking comparing to the rest of the population. Please clarify.

5. The section reporting incident cardiometabolic disease is relatively brief. More detailed results and discussion would be appreciated.

6. Various sensitivity analyses could strengthen the results, e.g., stratified analysis by age groups, repeating association testing considering only EHR data, testing for reverse causality.

7. For KM plots in Fig 4,

a) Please show the number at risk over time by strata.

b) Instead of survival probability, it may be more informative to present cumulative incidence risk.

c) Please include 95% confidence intervals for all curves.

d) In line 227, authors state that "Survival curves stratified by age and PpgAge gap nearly overlap, as well — the survival for a chronological 55 year old with a PpgAge gap of +6 is similar to a chronological 65 year old with a PpgAge gap of (slightly less than) -6." However, Fig 4b is an unadjusted KM curve. Does this hold even after adjusting for covariates? Please include adjusted risk curves.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

The authors satisfactorily addressed all the comments raised by implementing the paper or by providing a convincing response in the rebuttal letter. The manuscript is improved after the round of revision and I recommend to accept the paper.

(Remarks on code availability)

Reviewer #4

(Remarks to the Author)

Thank you for addressing my comments in detail - I learnt much from your response. In this report I only focus on one comment which I think may warrant further revision.

Original Comment 1: There is little analysis of how Ppg-Age changes within an individual over time (i.e. whether it increases by an average of 1 year per chronological year), and I thought this could have greatly enhanced the study. Furthermore, I envisage this would be feasible given that some participants contributed data for four years.

Thank you for including the analysis of PpgAge changes over time (Supplemental Figure S1). In your rebuttal you state that "these age deltas are better calibrated for younger participants". I would suggest going further than this by: (i) including discussion of this in the manuscript (I didn't see any discussion of Figure S1 (d), although perhaps I have missed it); and (ii) commenting on what this tells us about this plot tells us about the potential utility of PpgAge. Personally, I think that if PpgAge rate is not significantly higher than zero for particular age groups (e.g. age 45+ in females, and 55+ in males), then it raises questions about the ability of PpgAge to track changes in healthy ageing in these groups. In addition, if PpgAge rate differs significantly between age groups, then perhaps this tells us that the potential applications of PpgAge may differ between age groups (e.g. tracking healthy ageing in younger age groups, vs. stratifying according to disease risk in older age groups). I think some discussion of this would be helpful.

(Remarks on code availability)

n/a - no code provided

Version 3:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Thank you for thoroughly addressing the comments and improving the manuscript's clarity. The authors have sufficiently evaluated and responded to my comments.

(Remarks on code availability)

In the code provided, 'analysis/generate_synthetic_data.py' mentioned in README file is missing.

Reviewer #4

(Remarks to the Author)

Thank you for addressing my comments.

(Remarks on code availability)

My apologies, I don't know why I haven't looked at the code until now.

Please note that whilst I have had a brief look at the code, I have not reviewed it in-depth, partly because I'm not an expert at Python (making it difficult to provide a well-informed critique), and partly because I'm not an expert in all the packages and statistics used. I have also only reviewed the python code, and I have not reviewed the R-code (I've barely ever used R).

I note that as far as I can tell the code does not allow one to derive PpgAge from a PPG pulse wave - rather it allows one to perform the statistical analysis after PpgAge has been derived. Furthermore, it doesn't not allow one to recreate the original results, rather it uses synthetic data.

I have the following comments:

- The code appeared well structured and accompanied by a helpful readme file.
- I was able to get the code to run up until a point (I encountered an error in analysis_3_pregnancy.py)
- Might it be helpful to include a requirements.txt file? I didn't immediately see a list of required packages.
- The readme states that you can generate synthetic dataframes using 'analysis/generate_synthetic_data.py' . I didn't find this python file. However, I presume that I did generate synthetic dataframes, as I was able to create some csvs and plots.
- Generally, it is helpful to be able to follow the statistical analysis, even if only using synthetic data. However, it would be even more helpful if one was able to reproduce the original analysis, including not only the statistical analysis, but also the derivation of PpgAge from PPG pulse waves. Of course, I acknowledge that there are potential difficulties in facilitating this, e.g. (i) providing code for derivation of PpgAge may not be possible because of IP restrictions; and (ii) providing the original data is subject to the requestor going through the (lengthy) process outlined in the 'Data Availability' statement. In my view, this is simply a limitation of the ability of reviewers to critique the work (a limitation which is often, although not always, encountered in similar research). I would be happy for the manuscript to be accepted despite this limitation.

Version 4:

Reviewer comments:

Reviewer #4

(Remarks to the Author)

Thank you very much for tidying up the code. The README file reads well.

I have tried following the code, and wasn't able to. My apologies - I suspect this demonstrates my lack of familiarity with Python, rather than issues with the code. Therefore, I would suggest the following to the Editors:

- If you want the code reviewed, then I would suggest that someone with a good working knowledge of Python could check to see whether the code can be run. This could potentially be someone without knowledge of the research topic. I suspect that the errors I've encountered might be overcome by someone skilled in Python programming. If you want my input then I will be happy to provide input where possible (acknowledging that I am not an expert in Python).
- You may also decide that the code review has gone sufficiently far, and that you can make a decision without requiring further review.

In case it's helpful, the issues I encountered were:

- I received errors when running 'pip install -r requirements.txt'. e.g. 'ERROR: Could not find a version that satisfies the requirement torch==2.4.1'. I tried installing each package manually (e.g. 'pip install torch'.
- I then received an error when running 'python -m training.main -c TrainingPpgSsl', e.g.

... torch/utils/data/_utils/signal_handling.py", line 66, in handler
_error_if_any_worker_fails()

RuntimeError: DataLoader worker (pid 19944) is killed by signal: Terminated: 15.

I suspect that someone skilled in the art of Python might be able to handle these errors. Unfortunately, I don't have the level of understanding to know whether these can be handled easily or not.

(Remarks on code availability)

(this is a copy of my comments to the Authors)

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```

```
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```

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Response to reviewers

Submission NCOMMS-24-37590-T

We thank the reviewers and editor for their thorough assessment of our work. Below is a point-by-point response to each reviewer comment. A few common themes emerged among reviewers, which we address below and in the revised manuscript. These include

- **PpgAge and predicting incident disease/events.** We now include an survival analysis that uses PpgAge gap at baseline to predict incident disease diagnosis and events, including a composite endpoint of coronary artery disease events, new diabetes diagnosis, new heart failure diagnosis, and new valvular disease diagnosis. We show that PpgAge is significantly associated with increased risk in each case, even when controlling for other major risk factors. We believe this additional analysis addresses both questions about the strength of the association between PpgAge when controlling for other risk factors, and suggests a potential pathway for clinical translation (see new **Figures 4 and S4**)
- **Interpretability of our model.** We now include a new figure that visualizes low and high PpgAge gap predictions for different chronological age groups. These plots visually depict some larger scale features that drive the PpgAge prediction, and comport with expected features of aging present in PPG (see new **Figure S8**).
- **Additional summaries of hardware and software used.** We include a more thorough breakdown of device types and operating systems across all participants in our study. Further, we importantly note that across all OS's and hardware, the fundamental PPG sensor mode remains unchanged (see new **Figures S12 and S13**).

REVIEWER #1 (REMARKS TO THE AUTHOR)

Summary: The article titled "A Wearable-Based Biomarker of Aging and Its Association With Disease and Behavior" implements a novel self-supervised learning (SSL) approach to predict chronological age utilizing continuously monitored PPG data collected from consumer wearables in a cohort of AHMS. Additionally, it assesses the associations between the estimated aging measure, denoted as PpgAge, and disease diagnoses, smoking, and sleep stages. Furthermore, it demonstrates a substantial increase in PpgAge during pregnancy in women from a longitudinal analysis. Overall, the paper provides insights into utilizing data from consumer wearables combined with DL approach to predict chronological age and show its associations with behavioral and physiological measures.

Strengths:

1 – SSL has gained increasing attention and shown potentials in recent studies, some of particularly those using longitudinal, continuous data from wearable devices. The article is interesting and innovative by applying this DL approach to a new modality that has not been previously explored in the context of biological age estimation.

2 – This investigation aids readers in comprehending and identifying the possibilities of consumer devices in studying longevity or aging. Given the growing adoption of consumer wearables, such investigation holds potential for health monitoring and clinical usage from various angles, if scientific rigor and validation is warranted.

We thank the reviewer for constructive comments, fully addressed below, and for clearly noticing both the methodological and health contributions of this work.

Weaknesses:

1 – It is interesting to know that PpgAge has a high accuracy of predicting chronological age. However, the paper lacks evaluation of PpgAge in predicting other age-related outcomes such as mortality, incident diseases, or functional outcomes. The authors should demonstrate the additional PpgAge's predictive ability with these outcome(s).

We agree that the paper lacks evaluation of PpgAge (and age gap) predicting outcomes such as mortality, as such data is not available to us. To address this, we include a new survival analysis in **Figure 4** that highlights PpgAge gap's ability to predict

incident disease diagnoses and events. We found that PpgAge at baseline is a significant predictor of a composite endpoint of coronary artery disease events (new diagnosis, heart attack, unstable angina, stent, or coronary bypass) even when controlling for age, sex, BMI, smoking status, VO2max, hypertension, diabetes, and high cholesterol diagnoses. A PpgAge gap of 6 years corresponds to a statistically significant hazard ratio of 1.40 (95% CI 1.26–1.55), a stronger effect size than a previous hypertension or diabetes diagnosis, and nearly as strong as a previous high cholesterol diagnosis. We observed similar results for a new heart failure diagnosis and valvular disease diagnosis (see **Figure S4**) and for a new diabetes diagnosis (see **Figure 4**). Unfortunately, we do not have any participant mortality data — though we agree that this would be the most compelling outcome to study and hope to do so in the future.

We also note that our submission does include an extensive analysis of age- and sex-adjusted associations between PpgAge gap and existing disease diagnoses (e.g., **Figure 3**), which can be interpreted as the metric's ability to predict current disease state. For example, among 45-55 year old women, a participant with an age gap >6 years is twice as likely (i.e., has twice the base rate) to be diagnosed with diabetes (see **Figure 3b**). While predicting future disease has a clear motivation, inferring a participant's current (potentially unknown) health state is also well-motivated both for validation of the approach as well as applications — for example, a screening tool could adjust base rates (and therefore risk) of disease by not just age, but by PpgAge gap. And in particular, for high prevalence (and under-diagnosed) metabolic diseases like hypertension and diabetes, the age gap metric is highly associated with previous diagnoses — which we summarize in **Figures 3, S2, and S3**.

2 – In the introduction, better explanation and motivation of using PPG and SSL in age prediction are needed. Also, please consider that different wearable modalities may reflect different aspects of aging, e.g., PPG may be more reflective to vascular aging.

This is a great suggestion — in the introduction we now explicitly motivate PPG as a signal that may target cardiovascular signatures of aging. We contrast PPG to other wearable activity data to predict age (e.g., accelerometer or steps), which may leverage behavioral changes associated with (or driven by) aging, but can also be confused by other social factors. For example, the pandemic presented an extreme shift in physical activity behavior that can sharply change biological age estimates from physical activity data (Prykov et al, 2021). A core benefit of using PPGs from a subject at rest is that it isolates cardiovascular function, yielding a clock that is more robust to changes in behavior. The rest PPG signal reflects not transient changes in user behavior, but their physiological consequences. We now include these points in the discussion.

Additionally, we now include a discussion of the challenges of using raw PPG signals (as opposed to summaries like HR and HRV), which motivates the use of more data-driven feature extractors, like SSL. This is particularly important given the healthy population sub-sampling strategy — SSL-based features are far more data efficient than models trained end-to-end on a small amount of data.

We include this additional context in the introduction now (please see the red line changes).

3 – In methods, detailed information on the detailed breakdowns of types of consumer wearables (i.e., Apple Watch) and software versions used to collect the PPG data. Are all participants used the same smartwatch model during the study follow-up? This information is crucial, particularly with consumer wearables where hardware/software may vary largely.

We now include information about the different Watch models and different OS versions incorporated in this study.

Supplemental Figure S12 includes an additional data summary, which details the distribution of participants and segments across all watch models (**Figure S12c**) and OS versions (**Figure S12d**), as well as the distribution of the number of unique watch models and OS versions per subject — the vast majority of subjects only have 1 watch model, but we often see more than one OS version. Regardless, it is important to note that all 60-second PPG measurements (i.e., segments) used in this analysis utilized the same mode of sensor operation, across all Watch models and OS versions represented in the data set — we now make this explicit in the “Data processing, representation learning, and modeling” subsection in the Methods section.

4 – Is there information on the comparison or ascertainment of the Apple Watch's PPG to the ground truth? Such validation is highly important when studying with wearables.

There have been a variety of studies comparing Apple Watch PPG-derived metrics (e.g., HR and HRV) to reference measurements (e.g., ECG chest strap) under various conditions (Shcherbina et al, 2017; Fuller et al, 2020; Khushhal et al, 2017; Hernando et al, 2018). Most relevant to our study is the PPG mode that measures HR/HRV in the background or during breath sessions, which generates RR-interval estimates. Hernando et al, 2018 compare the RR-intervals generated by Apple Watch in this mode to a reference device (a Polar H7) and find “very good reliability and agreement” among a variety of HRV estimates (including SDNN, RMSSD, and pNN50). Further, they found that Apple Watch-generated RR intervals reflected changes induced by mild mental stress.

Additional research has studied the accuracy of PPG-derived heart rate and energy expenditure in an alternative mode, during exercise. This mode does not generate RR-intervals. Nevertheless, the exercise mode HR has been found to be accurate in comparison to other consumer wearables during low and moderate activity (Shcherbina et al, 2017, Fuller et al, 2020; Khushhal et al, 2017).

Additionally, a recent white paper published by Apple (Apple, 2024) details watch performance relative to a ground truth ECG sensor in various modes of operation. The operation most relevant to our study is the “tachogram algorithm,” which uses green PPG to measure heart rate and heart rate variability. In this mode, 99.6% of all instantaneous pulse rates are within 5 bpm of ground truth in the validation cohort (n=240 subjects, 266k tachograms) for normal sinus rhythm.

We now mention the accuracy of PPG-derived HR and HRV and cite Hernando et al, 2018 and the recent white paper in the introduction.

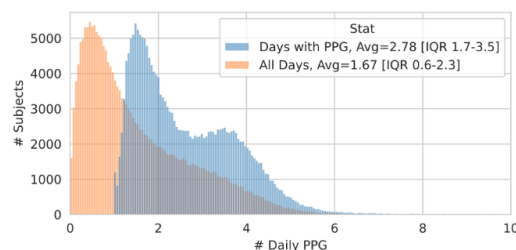
5 – Data processing lacks critical details on wear time, data quality, handling missing data, etc. Because SLL may mask or crop to remove parts, which can be problematic and may undermine abnormalities in case of insufficient monitoring period. Please provide a through explanation of the data preprocessing workflow.

As a rough surrogate for wear time, we now include an analysis and summary of the number of PPG segments observed per participant — see **Supplemental Figure S12a** (reproduced below). The number of background PPG segments sampled per participant vary by behavior and wear time as segments are opportunistically collected during detected periods of minimal movement. Due to the sheer amount of data collected, we are able to average over segments across days — for now we ignore time of day of each segment, but investigating within-day variability is a promising avenue for future work (e.g., morning vs day vs overnight segments).

Though these segments are collected with a variable frequency, over time we collect many samples per subject. We now include details on the PPG segment sampling and summary statistics on the frequency of background PPGs collected per participant in the Methods section and a reference to **Figure S12**.

Here, we summarize the distribution of daily PPG segments collected per subject per day in two ways:

- The per-subject average over the course of the whole study (orange); and
- The per-subject average on days with PPG collected (blue)



When PPG are collected, we see 2.75 segments collected on average.

Over the course of the study, we observe about 1.65 segments collected per day, on average. This includes, of course, days where the participant does not wear the watch. Indeed, there are days where we do not observe the PPG data, but on average (over ~200k participants) we collect at least 1.65 segments per day, and in some cases many more.

The self-supervised learning (SSL) algorithm that we use in this case does not use masking or removing parts — each PPG

measurement we use is a complete 1 minute segment sampled at 64Hz. We follow the data processing and SSL algorithm framework described in detail in [Abbaspourazad et al](#), which we cite in the Methods section. We only consider PPG segments that meet a quality threshold — a beat segmentation algorithm is able to find beats that cover at least 90% of the segment. Other quality metrics may work well, including limits on measured accelerometer magnitude over the entire segment, or signatures of the spectral decomposition of the signal. We have added details in the methods preprocessing section describing this filtering step.

6 — In the discussion, it is unclear how PpgAge could be interpreted or used in practical settings. It lacks discussion the explainability of model predictions, leaving questions about the practical implications of PpgAge and PpgAge gap. For example, if an individual is estimated with PpgAge gap of +2 years, thus “seemingly old”, what drives or constitutes this age gap value? What is the practical implication for this specific individual?

The short answer to these two questions — what makes someone “seemingly old” and what is its practical implication? — is that it’s not fully understood. And we note that this is the case with *any* aging clock (e.g., epigenetic, ‘omics, imaging, or PPG-based). Additional research is needed to show that slowing progression of *any* aging clock reliably corresponds to the slowing or arresting of age-related disease progression. And additional research is needed to show that certain interventions (e.g., lifestyle, pharmaceutical, etc) are able to reliably alter the trajectory of the aging clock. We note that all of these questions are being actively researched with the more mature and common set of aging clocks ([Moqri et al, 2023](#)).

At present, there is no real evidence-based decision to be made on the basis of a PpgAge gap of +2 years. However, we believe the first step is toward such insight is to establish (and study) the strength of the association between this new digital biomarker and disease — what we view as a major contribution of this work. In future work, we aim to study if taking action to reduce or slow the PpgAge progression reduces or slows the burden of age-related disease (perhaps specifically cardiometabolic or vascular disease). And it is here where we believe that PpgAge has the advantage of being a uniquely accessible and noninvasive clock, whereas epigenetic markers and lab-based clocks entail more friction to measure. Seeing a short term response (i.e., a reduction in PpgAge) over the course of a few months or years could indicate if an intervention is likely to reduce long term age-related burden.

To demonstrate potential utility beyond longevity research, we now include a survival analysis showing that PpgAge gap is a significant and independent predictor of incident cardiac events and disease diagnoses (as mentioned above). This analysis suggests that cardiovascular risk scores that incorporate such a gap could be more precise. Further, we have shown that PpgAge gap strongly correlates with disease prevalence in our study population, which suggests that PpgAge could be used in a variety of risk calculators (ie..g, modulating pre-test probability of hypertension, or peripheral artery disease).

Regarding explainability and what drives the prediction, we now include an interpretability analysis (see **Supplemental Figure S8**), visualizing segmented PPG beats that correspond to high and low age gaps across a range of chronological ages. We see that the age gap is picking up on known features of aging (see [Charlton et al, 2022](#) Figure 4) i.e., the diastolic notch and diastolic peak — prominent features of PPG waveforms — disappear as an individual appears older across all chronological age groups. While these features appear to be important to our prediction, it is possible that there are more subtle features with unknown physiological roots captured by the data-driven SSL approach.

Specific comments

1 — Title: I think the study’s central focus of the paper is chronological age prediction applying SSL to PPG data. ‘wearable-based biomarker of aging’ may be interpreted as an overstatement given the current results. Please consider revising the title to more accurately reflect the study.

We see the dual focus of this paper as (i) devising an accurate PPG-based normative age model, and (ii) understanding deviation of predicted age from chronological age (i.e., the age gap) and how it relates to disease and behavior. Beyond analysis of the accuracy of the model in various subpopulations, the main focus of analysis is the association of the PpgAge gap with disease and behavior (e.g., previous diagnoses, incident disease, physical activity, and smoking).

We note that studies of biological aging, including [Tian et al 2023](#) follow a similar procedure — predict chronological age from a

set of biomarkers in a healthy population — and we generally follow their usage of the term “biological age.” Though we recognize that the term “biological age” is overloaded, and even these references may also be viewed as an overstatement — as they are all built on statistical models predicting some age or aging surrogate.

In keeping with other similar references (e.g., [Ahadi et al, 2023](#)), we have edited the introduction to refer to the PpgAge biomarker as a PPG-based “aging clock.” We have also edited the title to reflect this object of study — “A Wearable-Based Aging Clock Associates with Disease and Behavior.” We also note that PPG is a ubiquitous health signal in consumer wearables, and our goal was to emphasize the wearable aspect of our work, as opposed to clinical pulse oximetry.

2 – Results

(i) For results presenting using quartile groups, please provide the exact cut-off values.

Thank you — we try to use absolute age gap cutoffs (in predicted years) as much as possible, but when percentile-based groups are used, we now report the cutoff values in captions or as in additional tables (e.g. **Table S1**, **Table S2**, **Figure S5c**).

(ii) What is the median (IQR) follow-up time? Please also include information on withdrawal or dropouts during follow up.

Of the original 213,976 participants that upload at least one PPG segment, the median follow-up time is 593 [IQR 292—996] days — this information is now included in **Figure 1d**. For analyzing dropouts in the study, a good reference for the Apple Heart and Movement Study is the recently published work [Truslow et al, 2024](#). In it, there are a few ways that follow-up time, withdrawal, and dropout are quantified: Figure 1 displays the fraction of participants that do not contribute any passive data (e.g., stand hours or research app data) as a function of days since enrollment, and Figure 2 displays the fraction of participants that responds to surveys in their first year of enrollment. Table 5 reveals that about 79% of participants complete the baseline medical history survey.

In our cohort (which includes more participants than the analysis in [Truslow et al, 2024](#)), in addition to **Figure 1d** distribution of days in the study, we also now include a plot of the fraction actively contributing PPG as a function of number of days in study, in supplemental **Figure S12**. The trends are similar to those identified in [Truslow et al, 2024](#).

(iii) What are the cohort characteristics? Provide a table including baseline characteristics.

We supply a table of cohort baseline statistics in **Figure 1b**, including a breakdown of subjects in the general and the “healthy” cohorts by biological sex, including age, BMI, (estimated) VO2Max, number of subjects on medication, with smoking history, with daily smoking behavior, and with a diabetes, hypertension, heart disease, or heart attack historical diagnosis. We refer to this table in the first paragraph of the results section. We now make it clear in the manuscript that these are baseline characteristics.

(iv) PpgAge gap seems not normally distributed. It would be informative to investigate why and include discussion.

Supplemental Figure S1 now presents a more thorough analysis of age prediction errors, including a QQ plot depicting deviations from normality. Within the healthy test cohort, we see that the deviation from normality is most influenced by one to two outliers, particularly in the high age gap direction. One potential explanation is that the healthy labels for these outliers were noisy or incorrect (recall that this is an observational study on a large, self-reporting population). In the general cohort, non-normality is likely driven by both disease, and the inability of the model to reliably extrapolate to the tails of the test distribution (i.e., a distribution shift). Another possibility is mis-reporting of ages. Please also see Bland-Altman analysis requested by Reviewer 4 in Supplemental **Figure S1c**.

(v) Please consider whether employment, shift work, and chronotype impact the associations in sleep analysis. For example, present results on the employment type, shift work status, or chronotype. And/or consider excluding shift workers from the analysis.

This is a great point — the nature of work schedule could certainly mediate the association between sleep and PpgAge gap.

Unfortunately, we do not collect fine grained employment data that would reliably represent shift work status. We now call this out as a limitation in the discussion section, and cite the literature that has established a firm relationship between shift work and metabolic disorders (e.g., [Karlsson et al 2001](#), [Wang et al, 2014](#)).

(vi) In the association of PpgAge and pregnancy,

a) Is the PpgAge increase at the same rate across different age groups of mothers?

b) Please consider sensitivity analyses stratifying by/controlling for healthy vs. pre-condition status, BMI, comorbidities, socioeconomic status, and demographic factors.

We now break down the analysis into subgroups by (i) baseline age, (ii) baseline BMI, and (iii) healthy/non-healthy pre-condition status. We do note that our samples become quite small in this secondary analysis. In all three cases, we do not see a significant difference in the trend, though we do observe that within each subgroup considered, the trend still holds (a roughly 3 predicted year increase, peaking at parturition time). See additional plots in **Supplemental Figure S11**.

c) In pg. 6, “we observe a slightly negative median age gap slope for a year post parturition”, does this imply that childbirth is associated with a permanent increase in biological age for women, even if they are otherwise healthy and disease-free?

We do not claim that childbirth is associated with a permanent increase in PpgAge, even if otherwise healthy and disease-free.

The observation window postpartum is short, so it is unknown whether PpgAge reverts to pre-pregnancy baselines (adjusted for time passed), but we do see a flattened trajectory relative to non-pregnancy. **Supplemental Figure S11a** includes a much longer post parturition time window, but drastically reduces the sample size of available participants (from n=165 to n=29). Over that year, we see a slight median decrease — over this time one would expect an increase of about a year for individuals in this chronological age band (see new **Supplemental Figure S1d**).

The overall goal of this analysis is not to learn something new about pregnancy, but to leverage the physiological changes due to pregnancy to study PpgAge. We perform this analysis because of the apparent sensitivity of epigenetic clocks to pregnancy — e.g., PCPhenoAge (+2.39 years), PCGrimAge (+1.19 years), GrimAge2 (+2.52 years) ([Pham et al, 2024](#)). We thought it was notable and worth reporting that the PpgAge clock exhibits similar (though not exactly the same) behavior, where the pregnancy period is associated with about a +3 year increase.

(vii) Figure 4(a)-(b) captions do not seem to match the text. (pg. 4)

Thank you for pointing this out — fixed.

(viii) It is unclear how “exercise minutes” are quantified (pg. 5). Is this using MET or MVPA? Please provide detailed explanation.

Exercise minutes are quantified by Apple Watch via an algorithm which uses the accelerometer to estimate how many full minutes of movement equal or exceed the intensity of a brisk walk — an estimate of moderate to vigorous physical activity (MVPA). During workout sessions (actively logged by users), the watch uses heart rate and GPS in addition to accelerometer to count exercise minutes. It is not an explicitly METs-based quantity. The estimates are available via the [HealthKit API](#), and is summarized by the green ring on the Apple Watch interface. The rings are discussed in more detail in [this publication](#) on health nudges (which looks more closely at the stand or blue ring). We have added this description and link to the HealthKit API to page 6.

3 – Methods

(i) Please include more detailed information on the model evaluation e.g., data split for training and evaluation and internal validation.

(ii) Please clarify what packages, at what stage, were used in R and python.

(iii) External validation in another dataset is strongly encouraged.

For (i), we train the normative age model by using 80% (n=5,355) of the self-reported “healthy” cohort, and report prediction performance and test statistics using *only* the held out 20% (n=1,373) and the entirely held out “general” cohort (n=120,235). These are all linear models (using a ridge regression penalty) built upon the fixed SSL representations that are learned in a preprocessing step. The strength of the regularization is tuned using cross validation (as implemented in `sklearn`). Performance is not particularly sensitive to regularization. We have included these details in the Methods section.

For (ii), we use `pyspark` for data pre-processing and large scale data queries, `pytorch` for SSL pre-training, and `scikit-learn`, and `numpy` in `python` for training the age prediction models and computing confidence intervals (via the bootstrap), and `matplotlib`, `seaborn` (in Python) and `ggplot2` (in R) for visualization. We have added these details to the code availability statement.

For (iii), we agree that validation on an external dataset would strengthen this submission, but for this submission, such an external validation is infeasible. We do note the geographic diversity of our cohort, with participants from each state in the US, and in both urban and rural areas. However, we do recognize that it is one study cohort, and that is a limitation. We explicitly now call out this limitation in the discussion section. In future work, we will validate these estimates against external datasets.

REVIEWER #3 (REMARKS TO THE AUTHOR)

The authors took advantage of data collected with Apple watches from 200 thousands participants to estimate a measure of biological age (also called "aging clocks") through the waves recorded by photoplethysmography (PPG), elaborated using a machine learning approach and setting chronological age as the "output" variable to build the clock. The idea per se is really interesting, and, as reported by the authors, in line with the ongoing research efforts aimed at finding a measure of biological age able to implement risk stratification for common age-related diseases (and/or mortality) on top of canonical risk factors without expensive or complex analyses. Thus, the topic is of high interest and the general approach is meritorious. The manuscript is well-written. However, I have some doubts relative to some passages of the paper and, more broadly, on the basic idea itself. In detail:

We thank the reviewer for this fair summary — indeed it sounds like we are in agreement on motivation for this work. Below (and in response to other reviewers), we hope to assuage your doubts.

1- The authors quickly define the input measure (with a couple of lines in the introduction). Researchers in the field of aging or CVD might not know what is a photoplethysmography and how it works and especially what it's "sensing". It is stated that it detects changes in blood volumes but I don't think this is a precise definition. I guess it senses blood volumes in a very specific area (the microvascular bed) and not at the organismal level. Thus, it basically senses the movement of blood through small arteries in the arm. This is likely the result of a combination of heart rate and blood pressure (which reflects arterial elasticity and overall blood volume, among others). Thus, such measure is very intrinsically linked to the health of the cardiovascular system. As a result, it is not surprising that it performs very well in the Cross-sectional detection of CVD, but fails to detect other age-related diseases (e.g. cancer, as reported by the authors). This implies that such an approach might be better suited to detect cardiovascular aging, rather than overall organism-level, biological aging. Indeed, even though the geroscience hypothesis suggests that few, common aging pathways underlie the development of all age-related diseases, recent manuscripts in the field suggest that different organs might age at a different rate within the same person and among individuals (DOI: 10.1016/j.arr.2024.102253; <https://doi.org/10.1038/s41576-022-00511-7>). Thus, I suggest to: 1- better explain in the introduction what is measured by PPG and what determines such waves; 2- Tone down some of the statements relative to the ability of this tool to detect biological aging; 3- Discuss (in the discussion section) the possibility that this tool might works well with cardiovascular conditions but not with other age-related diseases and, more broadly, that such an approach is intrinsically linked to cardiovascular health.

These are great suggestions. We now include an additional paragraph in the introduction detailing PPG and motivating its use for measuring cardiovascular-related signatures of aging. Further, in our discussion of alternative biological age clocks, we discuss their ability to measure only a specific aspect of aging, as opposed to the wholistic, multidimensional (and heterogeneous) concept of "biological age." In our discussion motivating PPG, we specifically refer to "cardiovascular-related signatures of aging" as a potential mechanism. Please review these changes on page 2 in the introduction.

In the discussion section, we expand on the note that our understanding of the physiological mechanism is limited, but that given the modality of the signal we are using, a plausible prediction pathway is signatures of cardiovascular aging (e.g., arterial stiffness, reduced cardiac output). Please see changes in the introduction and discussion.

2- For the same reason, the authors tried to compare their approach with an alternative one taking advantage of heart rate and heart rate variability, which are measured by the watch through the same technology. Their approach outperformed the other in detecting chronological age but this aspect is not discussed. Why the chosen approach is superior? Which biological aspect is missed when using only heart rate? Arterial elasticity? Blood pressure? Please discuss briefly in the discussion section.

Heart rate and heart rate variability do convey information about cardiovascular fitness and autonomic nervous system function. As these measurements were taken opportunistically at rest, the heart rate measurement (particularly an average over 30 days of measurements) likely reflects a value similar to a resting heart rate. Lower resting heart rates tend to indicate high cardiorespiratory fitness (though, of course, there are many potential confounders) (Jensen et al, 2013). Heart rate variability is also an indicator of autonomic nervous system function and stress (Kim et al, 2018).

While these are two very informative physiological summaries, empirically we find that they are not nearly as correlated with the much more detailed morphological information available from the entire PPG waveform. Shapes in this waveform are known to vary in age-dependent ways, as arteries stiffen, myocardium thickens, and cardiac output wanes. We believe the ML model is identifying subtle aspects of these physiological changes, which more strongly correlate with chronological age in the healthy cohort (see visualizations in **Supplemental Figure S8**). We have added the above to the discussion section.

3- The authors used chronological age as the variable to build the clock. This approach has been often used but available data (from epigenetic, metabolomic, or proteomic-based clocks) suggest that this approach is very precise in estimating chronological age but has a poor performance in longitudinally identifying the incidence of age-related disease or mortality (DOI: 10.1016/j.arr.2024.102253; <https://doi.org/10.1038/s41576-022-00511-7>), which should be the final goal of metrics of biological age. The results presented indeed sustain a cross-sectional association with some diseases, but only a preliminary analysis is shown relatively to the longitudinal association with CVD (and pregnancy, which is not a disease). The data relatively to the longitudinal incidence of CVD should be better presented and explained. Which is the delta of years associated with an increased risk of CVD? Which are the relative hazard ratios? Does it predict CVD also when adjusting for cardiovascular risk factors (e.g. ldl cholesterol, blood pressure, HbA1c in case of diabetes) or it improves risk stratification of established models when estimated age is used instead of calendar age (or the delta is added as a further covariate)? Please perform and present additional analysis on this aspect.

Thank you for this reference. Indeed, an apparent trade-off between the precision of age estimates and the predictive power of age gap of age-related disease or mortality has been established (with the extreme of perfect age estimates as being no better than age itself, of course).

However, there is also evidence that selecting participants to learn a *normative* aging model — i.e., by training a model on only healthy participants — can produce an aging clock that is predictive of overall mortality (Tian et al, 2023). In our study, we investigate this hypothesis by comparing a normative aging model (i.e., trained only on healthy participants) to a general aging model (i.e., trained on a random subset of participants). And similarly, we found that while age estimates are slightly more precise on the general cohort when training on a random subset (as expected), the age gap association with disease diagnosis is weaker across nearly all diseases in all age/sex groups (see **Supplementary Figure S10**). This result is consistent with the prevailing understanding — predicting chronological age without considering information about participant health yields a less useful age clock. We have added this reference and the above to the discussion section.

We included the pregnancy analysis because of recent work that showed that epigenetic age clocks see a large acceleration during the pregnancy period (followed by a reversion post parturition) (Pham et al, 2024). Our investigation revealed similar (though not the same) behavior for PpgAge.

Regarding longitudinal analyses, we agree that these sections should be fleshed out. We now include a survival analysis of incident disease diagnosis and incident cardiovascular disease events in **Figure 4**. We found that PpgAge at baseline is a significant and independent predictor of adverse cardiovascular events — a PpgAge gap of over 5 years corresponds to an 30% increase in risk (on the same order as high cholesterol), when controlling for age, sex, BMI, smoking status, and hypertension, diabetes, and high cholesterol diagnoses. Similarly, PpgAge gap is a significant predictor of incident diabetes and heart failure diagnoses, when controlling for relevant risk factors. Unfortunately, we do not have any participant mortality data — though we agree that this the most compelling outcome and hope to study it in the future.

In addition, it is unclear why a measure that should predict long-term outcomes (i.e. biological age and age related diseases) should change in the close proximity of the event (the authors evaluated the 180 days before a CVD event). This is not in line with the idea of biological age and with the mentioned geroscience hypothesis. Indeed, biological age should have a long-term performance, in order to allow the application of preventive strategies (e.g. statins, anti-platelet drugs, exercise, dieting) in patients at high risk that were classified not at high risk through conventional risk factors (e.g. blood pressure and LDL). If the measure changes in the close proximity of the event, it gives the impression of something closely linked to the condition but not really measuring the biological age of the individual. Also, what if a larger period of time is considered to estimate the measure of age or the delta (e.g. 2 years of measures instead of 180 days)? The performance in predicting CVD is the same?

Please present the results with more granularity and discuss accordingly.

Respectfully, to flip the argument around, it is unclear why a measure that should predict long-term outcomes *shouldn't* change in close proximity of a major disease-related event. If the final goal of metrics like “biological age” is to “longitudinally identify the incidence of age-related diseases or mortality” as you state above, then surely a major disease-related event — e.g., a heart attack — would represent an abrupt change-point in an individual’s overall risk of mortality. A history of heart attack predicts future heart attack and mortality. Because our aging clock is likely rooted in signatures of cardiovascular function related to age (as measured by PPG), it is not unexpected that large changes in cardiovascular function (due to disease-related events) induce large changes in age clock predictions. Being sensitive to short term physiological changes does not preclude “long-term performance to allow the application of preventive (sic) strategies.” Indeed, other age clocks show short term sensitivity to physiological changes that are not intuitively thought of as “aging” — see, e.g., pregnancy and stress (Pham et al, 2024 and Poganik et al, 2023). In fact, we view it as a positive feature that PpgAge is showing signs of longitudinal sensitivity to disease related events, and that the trend is in the intuitively correct direction (i.e., disease related events → older looking).

Other aging clocks (e.g., epigenetic) show short term sensitivity to physiological changes (e.g., pregnancy) that may be unexpected, particularly for the theoretical construct “biological age.” The results from Pham et al, 2024 and Poganik et al, 2023 demonstrate that short term stressors can accelerate DNA methylation-based “biological age” measures — including DNAmPhenoAge, DNAmGrimAge, and DunedinPACE — in response to some trauma surgeries, pregnancy, and COVID-19 infection. These popular aging clocks — whose measurements themselves are often referred to as “biological age” — all yield this theoretically contradictory behavior under the assumption that “biological age” should not be responsive in the short term to major health events or physiological changes.

Further, it is conceptually difficult to precisely delineate “something closely linked to the condition” and the “biological age” of an individual. Would arterial stiffening be closely linked to biological aging but not closely linked to hypertension?

Like all other aging clocks, what we propose is a statistical summary of a biological assay that is converted into units of predicted years. More specifically, PpgAge identifies statistical patterns in wearable PPG waveforms (and therefore, probably signatures of cardiovascular function) that highly correlate with age in self-reported healthy participants. In this sense, we have constructed a normative aging model from wearable PPG, which we now refer to as an “aging clock” in the updated manuscript (not “biological age”). Like all other aging clocks (e.g., Horvath 2013, Lu et al, 2019, Tian 2023, etc) PpgAge is a statistical summary of observables, not a ground truth reproduction of an individual’s theoretical “biological age” — a concept that still evades an agreed upon definition (Cohen et al, 2020). There is no guarantee that the behavior of such a metric would cleanly map to the expected behavior of the idealized biological age quantity (e.g., monotonicity, or insensitivity to disease-related events). Such shortcomings apply to all aging clocks, and it seems that we have further contributed to this confusion by referring to PpgAge as a “biological age” estimate. We apologize, and have attempted to be as clear as possible in the current manuscript about what we have proposed, how it relates to the theoretical “biological age” concept, and how it may deviate. Note that the revised manuscript now includes additional details in the introduction and discussion, as well as new experiments and visualization, to further support and clarify these concepts. We believe the inconsistency can be somewhat resolved by conceptualizing aging clocks an imperfect “estimator” of a theoretical “estimand”, in statistics parlance.

Regarding longitudinal measures, in our study, we looked at shorter term (i.e., 180-360 day) trends as a matter of convenience — expanding the observation window reduces the available sample size to study. Our goal for this analysis was to see if short term variability around events that may remodel the cardiovascular system (e.g., surgeries, heart attack) are associated with longitudinal changes in PpgAge — almost as a way to partially explain a physiological basis for this signal. Indeed — we would be surprised to see no change in one’s PpgAge trajectory in the face of such major events, given what we know about the PPG signal. We thank the reviewer for this stimulating comment, and we hope this response, complemented by the new discussions and results in the revised manuscript, are satisfactory to at least partially address it.

4- Relatively to the results in type 2 diabetes (T2D), it is known that this condition increases the glycation of protein and promotes the accumulation of advanced glycation end products. In turn, AGE accumulation is associated with an alteration of skin autofluorescence (SAF) which can be measured by specific instruments and correlates with multiple cardiovascular conditions in T2D. In there the possibility that the optical reader of PPG is affected by this phenomenon? This would not

represent an issue per se but, again, it would link the approach to a specific condition. If my question is appropriate, please discuss. If not (I'm not an expert of PPG), please ignore my suggestion.

We believe it is unlikely that AGE fluorescence would have a significant direct optical effect on 525 nm PPG. Skin autofluorescence is a very weak signal, and is not excited by green light. Further, the dark subtraction step in our PPG processing subtracts out background light. A reliable estimate of AGEs would require excitation and measurement at multiple wavelengths. A well-cited resource for noninvasive AGE estimation uses an excitation source of 300-420 nm and measures light from the skin between 300-600nm to produce an estimate of AGE concentration ([Meerwaldt et al, 2004](#)).

Based on our above interpretation, we speculate that it is unlikely that skin autofluorescence due to AGEs is a major driver of the PpgAge prediction and the source of its strong association with diabetes status.

REVIEWER #4 (REMARKS TO THE AUTHOR)

In summary, may I thank the authors and their institutions for this manuscript on photoplethysmography-based assessment of biological aging, which I have no doubt will make a significant contribution to the field. In this report I highlight what I perceive to be the key strengths and weaknesses of the study, and provide some recommendations of areas for improvement. Please note that I have some experience in photoplethysmogram (PPG) signal processing, but less experience in statistical analyses, so my comments are predominantly focused on the PPG-related aspects of the manuscript.

We thank the reviewer for the kind words and thorough review. A point by point response follows.

KEY RESULTS

The authors present a large-scale study of 'Ppg-Age', a parameter derived from wrist photoplethysmography signals which is designed to predict chronological age. Ppg-Age is found to correlate well with chronological age. In addition, higher a 'Ppg-Age gap' (i.e. Ppg-Age minus chronological age) is found to be associated with: (i) higher rates of historical (rather than future) heart disease and diabetes diagnoses; (ii) lifestyle behaviours (namely smoking status, "exercise minutes", and sleep characteristics. Finally, 'Ppg-Age' appears to increase more rapidly in the latter stages of pregnancy than either early in pregnancy or soon after giving birth.

Discussed in this document and summarized in **Figure 4** is a new key result, which finds that elevated PpgAge gap is significant predictor of future major adverse cardiac events, diabetes diagnosis, and heart failure diagnosis, even when adjusting for relevant risk factors (including chronological age, sex, BMI, smoking status, and chronic disease diagnoses).

VALIDITY

There are limitations to the quality of the study data, including:

- PPG signals were acquired using reflectance photoplethysmography and green light using a device with potentially variable contact pressure and shielding from ambient light. I don't consider this to be a substantial limitation, as "dark subtraction" was used to attempt to mitigate against the effects of ambient light, and the performance reported will reflect (no pun intended) the limitations of the pulse wave sensor.

This is a great point, and a potential source of variability that we now discuss. In fact, this kind of variability (e.g., variable contact pressure and ambient light effects) motivates the use of the self-supervised learning (SSL) pre-training step. Intuitively, the goal of the SSL loss is to only retain features that are important to distinguish individual subjects from one another, and jettison the rest. The extent to which these sources of variability — e.g., variable contact pressure and ambient light effects (that remain after dark subtraction and filtering) — reliably differentiate an individual participant from another randomly chosen participant will determine how much of this variability is retained in the SSL representation. The final prediction step — a linear model that maps these representations to an age prediction — will further jettison uncorrelated variability. Note that the revised manuscript incorporates additional details on the learned PPG representation.

And empirically, despite these sources of nuisance variability, our chronological age predictions are quite accurate on this naturalistic, free living data. We would argue that our prediction accuracy has demonstrated that our algorithmic approach has largely overcome such sources of nuisance variability.

- Several outcomes were obtained from self-reported data, which may introduce inaccuracies. This was noted by the authors, and would likely reduce the ability of Ppg-Age to predict chronological age, and reduce the strengths of the associations with outcomes. Therefore, I do not see this as a substantial limitation.

Absolutely, we are aware of and attempt to be forthright about this limitation. Further, if we expect that disease diagnoses are more likely to be under-reported (as opposed to over-reported) the association between PpgAge gap and disease history would likely be stronger than it appears in the statistics. We have included this summary in the discussion. Further, we have included a new analysis in **Supplemental Figure S1** of the normality of PpgAge gap, showing some outliers in the healthy test cohort, suggesting perhaps a handful of inaccurate labels or underreported medical history.

- Little physiological interpretation of 'Ppg-Age' is provided (besides mentioning that "The full PPG waveform reflects changes in blood volume in the microvascular bed of tissue"). This is acknowledged by the authors, and again does not in my view represent a substantial limitation.

We now include **Supplemental Figure S8**, which visualizes characteristic PPG beats within a cross section of chronological and age gap categories. With this visualization, we include more physiological interpretation of the signal and the PpgAge biomarker. When visualizing segmented PPG beats for high and low age gaps across a range of chronological age, we see that the age gap model is picking up on the typical features of age (see [Charlton et al, 2022](#) Figure 4) i.e., the dicrotic notch and diastolic peak disappears as an individual appears older. While these features appear to be important to our prediction, it is possible that there are more subtle features with unknown physiological roots.

In the introduction, we now motivate PPG as a useful signal for assessing cardiovascular-related correlates of aging, including a discussion of expected changes in cardiovascular function with age.

- Comment 1: There is little analysis of how Ppg-Age changes within an individual over time (i.e. whether it increases by an average of 1 year per chronological year), and I thought this could have greatly enhanced the study. Furthermore, I envisage this would be feasible given that some participants contributed data for four years.

We now include statistics on PpgAge changes over time — see **Supplemental Figure S1**. We show the predicted age rate — $\frac{\Delta PpgAge}{\Delta Time}$ — for participants for which $\Delta Time$ is at least 2 years. We stratify by chronological age, sex, and healthy/general cohort. A major trend we see is that these age deltas are better calibrated for younger participants — a phenomenon possibly attributable to distribution shift, and one that we aim to study further in future work.

SIGNIFICANCE

In my view this study is highly significant in several regards. First, it continues the narrative that large-scale studies of physiology and clinical outcomes can be conducted using consumer wearables, leading to valuable insights. Second, it demonstrates that it is possible to predict chronological age relatively accurately from wrist PPG in a large cohort. Third, it identifies potential bases for using PPG-derived parameters of biological age (namely associations with disease diagnosis, associations with lifestyle behaviours, and changes in pregnancy).

We appreciate this open-minded interpretation of our study, and wholeheartedly agree!

DATA AND METHODOLOGY

I have already commented on the limitations of the study data. I think the study appears to be well-conducted, the manuscript is well-presented, and unfortunately I do not feel qualified to judge some of the statistical methodology. However, I would highlight the following:

- Comment 2: I think the use of 'mean absolute error' to quantify agreement between Ppg-Age and chronological age is useful, but that it could be supplemented by additional statistics to give greater understanding of the level of agreement. Whilst I acknowledge r , r^2 , and $rmse$ are already used, I think these also do not provide a complete view. I would recommend reporting the bias and limits of agreement (as proposed by Bland and Altman), to help understand the accuracy and precision of Ppg-Age.

We now include a more thorough residual analysis in **Figure S1**, including a Tukey/Bland-Altman plot (**Figure S1c**). This visualization makes clearer what we might expect given the distribution shift, on the “general” cohort, we see some systematic under-prediction on the older cohort (>70 on the X-axis) — this range was largely unobserved in training. We also present a QQ plot and histograms of the PpgAge gap (i.e., residual) compared to theoretical normal distributions. Within the healthy test cohort, deviation from normality appears to be most influenced by a small number of outliers, particularly in the high age gap regime. In the general cohort, non-normality is likely driven by disease and extrapolation error (i.e., a distribution shift).

- Comment 3: Training and Testing data: I was confused about what data were used in pre-training and training, and what data

were used to obtain the results. For instance, the manuscript states that "We pre-trained our PPG foundation model using 19,993,427 PPG segments from 172,318 participants". Please clarify whether some/all of these 172,318 participants are also used for training and/or to obtain the results. If so, please comment on the appropriateness of this (I am unfamiliar with whether such information leakage would be a limitation or not). Second, I get the impression that the model was trained on the "healthy" cohort, which in principle I see as acceptable as results are reported for the separate "general" cohort. However, in practice, the headline result appears to be "PpgAge predicts chronological age with high accuracy in both the "healthy" and "general" group", without an explicit acknowledgment at that point in the manuscript that the "healthy" cohort were used for training. Furthermore, the next sentence states "Prediction accuracy is highest in the healthy cohort" (in comparison to the general cohort). For this to be a fair comparison, presumably ideally it would be a comparison between healthy and general data which were not used for training (whereas in reality my understanding is that the healthy data were used in training, but the general data were not). Therefore: (i) if I have misunderstood, then please correct me and clarify the manuscript; or (ii) if I have understood that "healthy" cohort data were used for training, then please consider this when interpreting the results.

For pre-training, we use a sample of about 20m PPG segments from about 172k participants. This includes the majority of participants with at least 30 segments — some or all of them overlap with the training cohort of healthy participants used to fit the normative age model. The self-supervised learning preprocessing step never uses age information — one could think of this as a sort of dimensionality reduction step (e.g., akin to PCA) which precedes a supervised model fit. The age predictions are then reported on participants that are completely held out of the supervised model fitting procedure.

To answer the second question — the healthy cohort with at least 30 PPG segments in the first month includes 6,728 participants (see **Figure 1c**). We divide this cohort into two sets — call them the healthy-train cohort (5,355 participants, ~80%) and the healthy-test cohort (1,373 participants, ~20%). We then train the age model only on the healthy-train cohort. All prediction error results reported on the "healthy" cohort are really on the "healthy-test" cohort (e.g., 2.42 years MAE, **Figure 2a**). Prediction errors on the general cohort (e.g., 3.13 years MAE, **Figure 2c**) are also reported on held out test participants — all of the participants in the general cohort were held out. In this sense, they are comparable errors — all model error results are reported on held out test participants.

The discrepancy in performance comes from, broadly, the distribution shift — the training data is identical in distribution to the healthy cohort test data (by construction) but different from the general cohort test data (see **Figure 1** — the age distributions are quite different). This difference in distribution is driven by sampling on health status.

Apologies for the confusion — we refer to both the healthy-train and healthy-test cohorts as the "healthy cohort." We have clarified this usage in the methods and results sections.

- Comment 4: I thought a strength of the study was the idea that one could train a model using chronological age as a reference. Much previous work has focused on using references measurements which are reflective of the underlying ageing process (e.g. carotid-femoral pulse wave velocity), presumably based on the assumption that this would be more informative. However, the authors have developed a helpful model using chronological age alone for training. Perhaps you might consider whether to mention this aspect.

Yes, thank you for pointing this out. We have explicitly noted this point in the discussion section, with an additional reference ([Rutledge et al, 2022](#)). Further, see discussion with the reviewer above about tradeoffs between predictive age accuracy and association with disease and mortality.

- Comment 5: Could you provide details of the equipment and algorithm used in this study, so that Apple Watch users can determine whether or not their watch measures PpgAge using the same approach reported in this study (e.g. same PPG sensor, and same algorithm)?

Reviewer #1 asked a similar question — we now include a new figure that summarizes the different Watch models and OS versions incorporated in this study. Supplemental **Figure S12** visualizes the distribution of participants and segments across all watch models (**Figure S12c**) and OS versions (**Figure S12d**), as well as the distribution of the number of unique watch models and OS versions per subject — the vast majority of subjects only have 1 watch model, but we often see more than one OS

version. Regardless, it is important to note that all 60-second PPG measurements (segments) used in this analysis utilized the same mode of sensor operation, across all Watch models and OS versions represented in the data set. In all aspects of our algorithm, we pool data from all Watch model and OS versions together. We also want to clarify that PpgAge is not a feature and is unavailable to Apple Watch users.

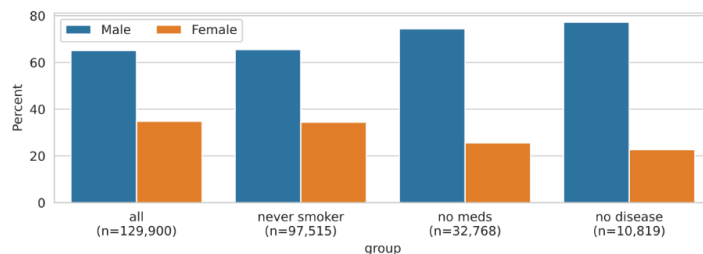
- Comment 6: Referring to Fig. 1(b), why are there: (i) fewer females in the 'All subjects' group (approx. 36% female); and (ii) substantially fewer females in the 'Healthy subjects' group (approx. 20%). Here, I think it would be helpful to provide insights into why: (i) more males than females enrolled in the study; and (ii) why the criteria for inclusion in the 'healthy subjects' group heightened this disparity.

Thank you for catching and pointing out this subtle detail. The Apple Heart and Movement Study is a naturalistic cohort that is not generally representative of the broader US population — the initial cohort consisting of about 36% female participants is a consequence of the observational nature of the study (see the [Truslow et al, 2024](#)). In light of this, we do generally present results by biological sex and age groups to control for these potentially non-representative demographic factors.

Why the biological sex disparity widens as we select a “healthy” cohort is a very interesting question that we explore here with an additional analysis (which is now included in **Supplemental Figure S13**). How we delineate a “healthy” participant from a “general” participant relies on three survey-based criteria

- No history of smoking (never smokers)
- No history of surveyed chronic disease (active response of “no”) (no disease history)
- Not taking medications (no medications)

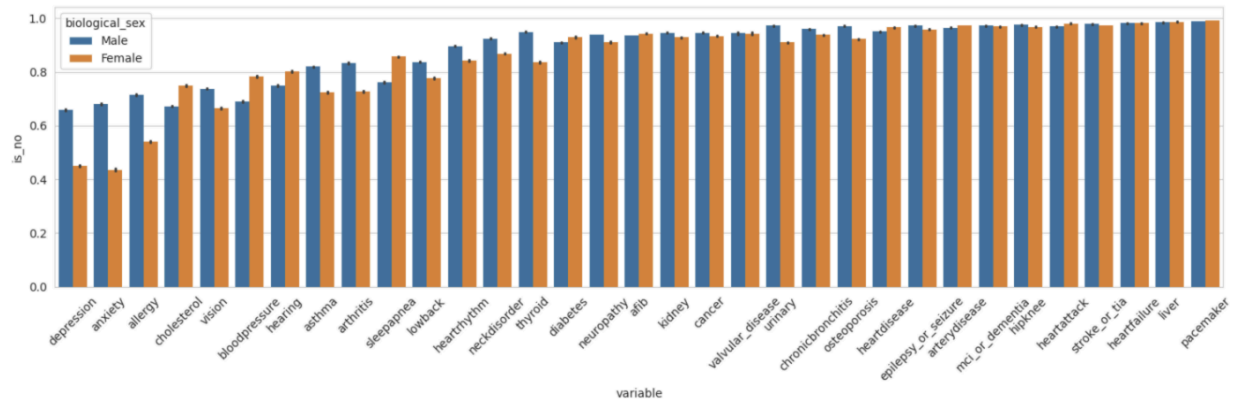
Looking at the original cohort of 129,900 participants (participants with at least 30 PPG segments observed in the first month between ages 18 and 85), roughly 35% are female, 77,905/213,976) we investigate the breakdown of each of these categories independently:



We found that roughly 36% of never smoker participants are female, roughly 26% of no medications participants are female, and 23% of no disease history participants are female.

Here we see the “never smoking” criterion does not affect the biological sex disparity of our cohort. However the disease history has a large effect — shrinking the proportion of female participants from 36% to 24%. The no medications criterion also has a large effect on biological sex proportion, but not as large as the no disease history criterion. All together, these criteria result in a cohort that includes 1,838 of 9,170 female participants (~20%).

Rates of self-reported disease and medication usage are higher among female participants. Seeing the large influence of disease history, we look at disparities in individual diseases. We found a major disparity in rates of depression and anxiety diagnoses among men and women, as well as smaller disparities for other conditions (e.g., high blood pressure, high cholesterol, asthma).



We investigated this ratio — a nearly 2 to 1 ratio of depression and anxiety diagnoses for women compared to men — in the literature, and indeed it largely matches a wide range of existing findings ([Kuehner, 2017](#)) as well as a quick examination of the CDC BRFSS survey data that asks this same question ([BRFSS Prevalence & Trends Data](#)).

This analysis raises an interesting question — what is the “best” way to define a normative aging model? We note that biological age is indeed associated with depression and anxiety in the test population — an association that is actually *strengthened* in the female population by subsetting to the healthy cohort (see **Figure 9b**), that is, by using an even less balanced training set.

For medications, we see the disparities shrink with age. For disease diagnoses, however, it does appear that women have a historical diagnosis at higher rates, for nearly all age categories. We summarize these findings in **Supplemental Figure S13**.

- Comment 7: I am sceptical of the comparison between Ppg-Age and the comparator "heart rate and heart rate variability" model (see Results section titled 'PpgAge predictions use morphological features beyond heart rate and heart rate variability'). This is because the heart rate variability (HRV) model only included two HRV metrics, whereas many more metrics have been proposed in the literature to quantify HRV. Because of this, could it be that the conclusion that "HR/HRV-based models predict age less accurately than PpgAge across all subpopulations" is partly because the model only incorporates a small subset of the potential HRV parameters, and perhaps the difference in performance between Ppg-Age and HRV would be less if the HRV model included more parameters?

The HR/HRV model comparisons currently use statistics that summarize the 30 days of PPG segments for each participant.

These include mean and standard deviation of heart rate, as well as mean and standard deviation of heart rate variability as measured by RMSSD and SDNN. We have fleshed out this experiment more, incorporating additional HRV measurements, including pNN50, pNN25, pNN10, and pNN5. Additionally, we add more than just mean and standard deviation statistics — we now include the min and max values of each statistic. See updated **Supplemental Figure S9**.

It's certainly not an exhaustive and completely conclusive experiment — perhaps there exists a more informative HRV measurement that totally explains why PPG can predict age. However, we believe this analysis, our new interpretability analysis in **Supplemental Figure S9**, and fundamental physiological understanding of the PPG signal suggest that other morphological features inform age predictions.

We discuss these new experiment results and the above interpretation in the discussion section.

- Comment 8: In the analysis of associations between behaviours and Ppg-Age, it appears that some investigations (e.g. sleep, exercise behaviour, disease) involve matching participants (e.g. by age and sex) whereas the investigation into smoking does not. Is this the case? If so, why not do matching for smoking status? If not, please clarify the methodology in the manuscript. Also, is it sufficient to match by age and sex, or would it be helpful to also match by blood pressure given its impact on Ppg-Age?

Apologies if this was unclear, but the smoking analysis summarized in **Figure 5** does decompose all subjects by age group (along the X-axis) and biological sex for male (**Figure 5a**) and female (**Figure 5b**) participants.

In the analyses presented in **Figures 2 and 3**, we are only examining marginal associations — that is pairwise associations between PpgAge and some behavior or disease of interest, controlling for biological sex and chronological age. We do recognize that further adjusting for behaviors, like smoking status, would shed light on the informativeness of PpgAge above and beyond other, more standard biomarkers.

Our new incident diagnosis analysis directly addresses this concern, summarized in **Figure 4**. As described above, these models control for age, sex, BMI, smoking status, and hypertension, diabetes, and high cholesterol diagnoses. We still see a significant association between baseline PpgAge gap and incident cardiovascular events.

- Comment 9: Please state the wavelength of green light used in the photoplethysmography sensor.

The wavelength of the green light is 525 nm — now stated in the Methods section.

SUGGESTED IMPROVEMENTS

- Comment 10: I would suggest further discussing the potential physiological basis for Ppg-Age, including the authors' perspectives on: (i) whether there is need to understand the physiology in order to use the parameter (notwithstanding their current remarks on "physiological mechanisms" and "longevity interventions"); and if so, (ii) potential directions for future research to further understand the physiological basis for PPG-derived markers of ageing (particularly reflectance PPG at the wrist). I would also suggest discussing the potential influence of blood pressure (see Fig. S2), and the (possible) benefit to understanding whether Ppg-Age provides additional information beyond that of blood pressure alone.

These are fantastic points for discussion.

We do not believe it is necessary to fully understand the physiology behind a measurement for it to be useful, but it is necessary to obtain solid evidence (i.e., prospective, randomized) that such a measurement is reliable and behaves as expected. For example, the mechanism behind other epigenetic aging clocks (e.g., GrimAge) is not fully understood, but some are still being studied as a shorter term surrogate endpoint to assess various health and lifestyle interventions ([Fahy et al, 2019](#), [Moqri et al, 2023](#)).

We do believe a better understanding of the mechanism is an important research direction, and could yield directions for new interventions. Some promising avenues for understanding the mechanism in reflectance PPG at the wrist include hybrid machine learning and physiological models (e.g., extensions of the model proposed in [Charlton et al, 2019](#)). Such an approach may be able to separate signal into the part of the prediction that is attributable to, say, arterial elasticity vs. cardiac output vs. a feature of unknown/unmodeled origin.

Regarding blood pressure — that is an interesting question. One might imagine that using average cuff measurements over a 30 day period would not predict age with such high accuracy. However — if we repeat the same SSL strategy with more fine-grained blood pressure measurements — e.g., the waveform from an arterial blood pressure measurement — it would not be surprising if a model built on such data would perform as well or better than our PPG age. This is, of course, speculation.

Relatedly, we now include a survival analysis that predicts time to major adverse cardiac events (**Figure 4**). In this analysis, we control for age, sex, smoking behavior, and a variety of other health conditions, including a diagnosis of high cholesterol, high blood pressure, or diabetes. Controlling for these other factors, the PpgAge gap remains a significant predictor of future adverse cardiac events — neither high blood pressure, diabetes, nor cholesterol explain away the effect of PpgAge gap. An age gap of 5 years at baseline corresponds to a 30% increase in risk in major adverse cardiac event. This level of risk is of a similar magnitude (and independent of) the effect attributable to a previous diagnosis of hypertension. We have added this in the discussion of the new survival analysis.

- Comment 11: Might the authors consider providing further suggestions as to the potential applications of Ppg-Age? Currently the Discussion suggests "effective longevity interventions" and "potential for clinical translation". It might be helpful to the reader to explore this a bit further, and in particular, consider whether any Ppg-Age functionality would need to be certified for medical use to be used clinically, and whether there is an expectation that consumers could respond to the Ppg-Age reported from their watch without clinical input.

The potential applications or clinical translation of any aging clock (e.g., epigenetic or other 'omics-based) may also apply to PpgAge. To be an effective surrogate for studying various longevity interventions, one must answer the question — does slowing or reducing predicted age reliably lead to the slowing or arresting of age related disease progression? That remains an open question even for the more mature aging clocks, and is subject of recent research ([Moqri et al, 2023](#)).

Further, how individual interventions (e.g., diet, exercise, pharmaceuticals) causally affect the surrogate PpgAge (or any other clock) is another open question that would need to be studied before clinical or lifestyle decisions are made on the basis of PpgAge trajectories. For example, it is still an open question whether something like [Oura's cardiovascular age](#) is an appropriate metric to target with lifestyle changes or medical interventions. We view this study (which began before Oura's announcement) as a first step toward building a practical and accurate wearables-based aging clock, and establishing a strong association between the new age gap metric with various physiological and behavioral conditions, including disease, pregnancy, exercise and smoking — a first step toward legitimizing these digital biomarkers.

It's possible that a future Ppg-Age health feature may need to go through a regulatory certification process to better understand the clinical actionability and to help users understand what the PpgAge means and how it's affected by their behaviors.

Regarding a non longevity-based application, we do show that the PpgAge gap is a significant predictor of future cardiac events. Though the evidence is retrospective and observational, our analysis suggests that it may be fruitful to incorporate PpgAge into cardiovascular disease risk models. These points have been added to the discussion.

CLARITY AND CONTEXT

The study is presented clearly, both in the text and figures, with a wealth of results presented in both the main text and supplementary material.

Thanks!

- Comment 12: The early stages of the manuscript (Abstract and Introduction) could give the impression that the assessment of biological age from the PPG is a novel idea. e.g. (i) "Existing measures of biological age are typically based on blood, saliva, vital signs, or clinical imaging. Wearables, however, can make convenient, frequent, and inexpensive measurements throughout daily life, scaled to an entire population."; (ii) the paragraph starting "Efforts to quantify biological aging" does not mention PPG-based approaches. I would recommend avoiding any possible impression that this idea is novel, because it is present in both the academic literature and wearable devices (e.g. Oura Ring's "cardiovascular age").

We thank you for this criticism — it was not our intention to ignore previous efforts using PPG to characterize signatures of aging (particularly predicting other biomarkers of vascular health as mentioned in the review linked below). Our dive into the literature led us toward these "aging clocks", and while we now cite various PPG and vascular age studies, we missed any PPG-based aging clock presenting itself as a tool for longevity research. Further we did (and do not) not see any wrist-based wearable PPG aging clocks in the literature. Oura ring's "cardiovascular age" was concurrent work that has yet to publish a peer reviewed (or any detailed scientific preprint) methodological description or study in relation to health, as far as we know.

We now include a reference to the mentioned review ([Charlton et al, 2022](#)) in the introduction, to make clear that studying the relationship between PPG and aging is not new. We try to stress that wearable wrist-PPG is our novel contribution, as well as the scale and varied analyses included in the study.

- Comment 13: I thought one key aspect of the context of the work was missing: there is minimal discussion of the prior art of assessing biological age from the PPG. This is a wide area of research (see our review - <https://doi.org/10.1152/ajpheart.00392.2021> for discussion of how wide this area of research is). A relatively novel aspect of the current study in comparison to previous work is that it uses widely monitored wrist-PPG signals, as opposed to finger-PPG signals (which I think more of the previous research has focused on). I would recommend discussing the prior art of PPG-based assessment of biological age in more detail.

Thanks for this reference and the insight. In our introduction, we now include a discussion of finger-PPG work focused on measuring vascular age as another motivation for this study. Additionally, we now include a discussion of the PPG-based biological age assessment, as well as a reference to your thorough review, in both the introduction and the discussion.

- Comment 14: I saw no acknowledgments or details of funding provided in the manuscript. Whilst perhaps these are provided in the submission system rather than the manuscript itself, I would like the reader to be aware of who funded the study, and any conflicts of interest.

Indeed we acknowledged funding in the submission system, but we have now added the following statements to the Competing Interests section (following references): “All authors are employees of Apple, Inc. and own Apple, Inc. stock. The Apple Heart & Movement Study receives funding from Apple, Inc. and the American Heart Association.”

- Comment 15: Fig 5: These images appeared highly pixelated on the PDF I was given. Please check that these and the other images in the manuscript are clear, and provided in a suitable format.

We have re-rendered these images as vector graphics — thank you for catching this!

REFERENCES

- Comment 16: It's unfortunate that Ref [App21], which is referred to in the context of the methodology used to estimate Vo2 max, refers to a webpage without a DOI. I wondered whether in light of this it might be helpful to provide more detail of the methodology used to estimate Vo2 max in the manuscript.

This is a good point, we now include a few more details from the linked white paper in the supplement.

Response to Reviewers

NCOMMS-24-37590A (4-4-2025)

We would like to thank everyone for taking time to review our updated manuscript. Point-by-point responses are below.

Reviewer #1 (Remarks to the Author):

Thank you for thoroughly addressing my comments. The additional information and analyses have clarified many initial questions. I have a few remaining comments and suggestions, mainly regarding the analyses of disease incidence, a key analysis to demonstrate the predictive accuracy of PpgAge.

1. Key details are missing in the analysis results:

a) Please report the total number of events (N events) for each disease.

b) When does person-time start and end in the analysis?

c) Over what time frame are these events assessed? Is this an “ever” diagnosis or within a specific follow-up period?

Thank you for taking a look at our response and new manuscript.

For (a), we now report the total number of events (N) for each disease in the results section, in addition to the existing counts in the table captions. We observe 901 diabetes events, 651 total CAD events, 212 heart failure events, and 265 valve disease events.

For (b), person-time starts when the participant begins at enrollment and lasts until either an event is present or observations are censored. Observations are censored either by the end of the study period or sustained survey non-response (i.e., the user “drops out” of the study).

For (c), we describe the median and IQR of time to event or time to censoring below. All of these are first events (e.g., new diabetes diagnosis, new heart failure diagnosis or exacerbation), that are observed within the study period (i.e., after enrollment, before dropout/censoring). All of this information is now detailed in the supplement, Section S4.

- Diabetes
 - Events: time-to-event median 353 days (IQR: 192 - 671)
 - Controls: time-to-censoring median 523 days (IQR: 294 - 833)
- CAD
 - Events: time-to-event median 479 days (IQR: 253-740)
 - Controls: time-to-censoring median 1059 days (IQR: 766-1280)
- HF
 - Events: time-to-event median 459 days (IQR: 277-722)
 - Controls: time-to-censoring median 1059 days (IQR: 763-1281)
- Valve disease
 - Events: time-to-event median 416 days (IQR: 198-681)
 - Controls: time-to-censoring median 1057 days (IQR: 761-1280)

2. In line 625, the authors state that disease diagnoses come from self-reports from repeated surveys and EHR.

a) What questions (and responses) and medical codes are used for disease diagnoses from self-reports and electronic records/Medicare, respectively?

Below is a list of codes and/or lab values used to label each condition. This information has also been added to the supplemental material, Section S4.

- (i) coronary artery disease (CAD) events: composite of MI, unstable angina, non-chronic angina, CABG, angioplasty, stent, new diagnosis of CAD (survey-only)
 - ICD10: I21.0%, I21.1%, I21.2%, I21.3%, I22.0, I22.1, I22.8, I22.9, I21.4, I22.2, I21, I22, I21.9, I21.A9, I201, I202, I208, I209, I240, I248, I249, I25111, I25112, I25118, I25119, I256, I25701, I25702, I25708, I25709, I25711, I25712, I25718, I25719, I25721, I25722, I25728, I25729, I25731, I25732, I25738, I25739, I25751, I25752, I25758, I25759, I25761, I25762, I25768, I25769, I25791, I25792, I25798, I25799.
 - ICD10-PCS: 0210%-0213%, 0270%-0273%, 02H03DZ, 02H13DZ, 02H23DZ, 02H33DZ, 02H00DZ, 02H04DZ, 02H10DZ, 02H14DZ, 02H20DZ, 02H24DZ, 02H30DZ, 02H34DZ
 - ICD9: 410.0%-410.9%, 00.66, 36.01, 36.02, 36.05, 411.1, 411.8, 411.81, 411.89, 413, 413.9
 - ICD9-PCS: 36.1%, 36.06, 36.07
 - SNOMED: 401303003, 703164000, 703165004, 15962541000119106, 15713081000119108, 15963181000119104, 44831000087103, 44841000087109, 840309000, 44851000087107, 44811000087108, 896691006, 44821000087100, 846683001, 15713121000119105, 868225002, 868226001, 868224003, 840609007, 840316004, 840312002, 868220007, 868214006, 868217004, 285981000119103, 846668006, 840680009, 285991000119100, 15712881000119105, 15712961000119108, 703213009, 703253007, 12238111000119106, 15712921000119103, 15713041000119103, 15713201000119105, 15712841000119100, 15713161000119100, 896697005, 896696001, 703211006, 703210007, 703209002, 401314000, 1204155000, 1204154001, 1204152002, 1204151009, 1204222000, 1208872002, 703360004, 1208873007, 57054005, 22298006, 4557003, 194823009, 15960741000119103, 15906661000119107, 15906061000119102, 15906461000119105, 15906541000119107, 15906341000119104, 232717009, 175021005, 736970002, 67166004, 8876004, 17073005, 10190003, 29819009, 14323007, 90487008, 1145199001, 1145198009, 1148690000, 1149059007, 1145200003, 1148691001, 1149060002, 736962007, 232719007, 736971003, 48431000, 736967001, 736963002, 359601003, 175048006, 175050003, 275215001, 275216000, 359597003, 119565001, 30670000, 119564002, 39724006, 870744008, 1010688007, 1217166005, 1217165009, 1217162007, 1017203005, 1010291002, 1217161000, 1017205003, 1217158001, 1217157006, 1217155003, 1217153005, 1217168006, 870743002, 1144644003, 1144647005, 1144889004, 1144646001, 1144645002, 1144890008, 871496000, 1017228002, 1010571005, 1017233003, 1017232008, 1017229005, 1010572003, 1017227007, 1010570006, 871497009, 868231004, 868230003, 1156984001, 1156983007, 1155686006, 1155687002, 868228000, 868227005, 232724005, 232720001, 736973000, 736965009, 74371005, 265481001, 175047001, 232721002, 736972005, 736964008, 10326007, 232722009, 39202005, 232723004, 82247006, 414088005, 418551006, 175029007, 871498004, 1209061001, 1209117007, 1209060000, 1209115004, 1208943002, 1208942007, 1208760004, 1208675002, 1209120004, 1208940004, 1208916007, 1208919000, 1208673009, 1208917003, 1209119005, 1208676001, 1209172009, 1208430005, 1204377006, 1204378001, 1209062008, 1209204000, 1208761000, 1209203006, 1209205004, 1201858000, 1209180002, 1209174005, 11101003, 75761004, 80762004, 91338001, 763725002, 175066001, 397193006, 397431004, 85053006, 36969009, 394659003, 413439005, 414795007, 413444003, 35461000087105, 35471000087101, 35491000087102, 35501000087105, 35511000087107, 35521000087104, 35541000087108, 35551000087106, 35561000087109, 35571000087100, 35481000087104, 35581000087103, 697976003, 428196007, 233823002, 194828000, 15960581000119102, 15960381000119109, 15960141000119102, 61490001, 41334000, 85284003, 89323001, 233821000, 4557003, 194823009, 15960741000119103, 371806006, 371809004, 371810009, 371812001, 371808007, 371811008, 315025001, 429559004
 - DRG: 280-285, 231-236, 246-251, 311
 - CPT: 92920-92944, 92980, 92981, 92982, 92984
 - HCPCS: C9600-C9608
- (ii) diabetes diagnosis, (a1c values from health records or self report new diagnoses)
 - Self-reported new diabetes diagnosis from monthly/quarterly changes in health survey

- HbA1c value of $\geq 6.5\%$
- (iii) heart failure diagnosis or acute exacerbation (we try not to use codes that indicate chronic diagnoses)
 - Self-reported new heart failure diagnosis from monthly/quarterly changes in health survey
 - A combination of ICD9/10 diagnosis codes, SNOMED codes, CPT procedure codes, HCPCS, and DRG codes
 - ICD10:
 - I501, I5020, I5021, I5023, I5030, I5031, I5033, I5040, I5041, I5043, I50810, I50811, I50813, I50814, I5082, I5083, I5084, I5089, I509, I110, I130, I132, I0981, I20.0, I25.110, I25.700, I25.710, I25.720, I25.730, I25.750, I25.760, I25.790
 - ICD10-PCS
 - 02HA% (except 02HA0YZ, 02HA3YZ, 02HA4YZ), 02WA0QZ, 02WA0RS, 02WA0RZ, 02WA3QZ, 02WA3RS, 02WA3RZ, 02WA4QZ, 02WA4RS, 02WA4RZ, 02WAXQZ, 02WAXRS, 02WAXRZ, 02YA0Z0, 02YA0Z1, 02YA0Z2
 - HCPCS:
 - M1150, M1151
 - ICD9
 - 39891, 40201, 40211, 40291, 40401, 40403, 40411, 40413, 40491, 40493, 428.0, 428.1, 428.20, 428.21, 428.23, 428.30, 428.31, 428.33, 428.40, 428.41, 428.43, 428.9, V4321, V4322
 - DRG:
 - 001, 002, 291, 292, 293, 215
 - SNOMED
 - 84114007, 56675007, 364006, 424404003, 195112003, 42343007, 10633002, 80479009, 698296002, 74960003, 16838951000119100, 194767001, 92506005, 101281000119107, 426263006, 426611007, 717840005, 15629591000119103, 67441000119101, 15629541000119106, 67431000119105, 23341000119109, 82523003, 96311000119109, 194779001, 15781000119107, 194781004, 5148006, 83105008, 43736008, 698594003, 195111005, 418304008, 153931000119109, 443343001, 153951000119103, 443344007, 120901000119108, 120891000119109, 120881000119106, 233924009, 898208007, 788950000, 446221000, 1204206001, 1204203009, 1204204003, 1204200007, 703272007, 703275009, 703273002, 703276005, 703274008, 10091002, 46113002, 85232009, 195114002, 71892000, 25544003, 44088000, 314206003, 367363000, 359617009, 1264003007, 80479009, 13839000, 44313006, 1208846006, 1208848007, 1208843003, 417996009, 443253003, 16253001, 120871000119108, 15629641000119107, 120861000119102, 15629741000119102, 120851000119104
 - CPT:
 - 33945, 33927, 33975, 33976, 33979, 33990, 33991
 - Left-ventricular ejection fraction $\leq 40\%$ from lab values in CHR
- (iv) valve disease diagnosis or valve replacement
 - Self-reported new valve replacement or valvular heart disease from monthly/quarterly changes in health survey
 - A combination of ICD9/10 diagnosis/procedure codes, SNOMED codes, CPT procedure codes, and DRG codes
 - ICD10:
 - I050, I051, I052, I058, I059, I060, I061, I062, I068, I069, I070, I071, I072, I078, I079, I080, I081, I082, I083, I088, I089, I091, I340, I341, I342, I3481, I3489, I349, I350, I351, I352, I358, I359, I360, I361, I362, I368, I369, I370, I371, I372, I378, I379, I511, I512
 - ICD10-PCS
 - 0027F04Z, 027F0DZ, 027F0ZZ, 027F34Z, 027F3DZ, 027F3ZZ, 027F44Z, 027F4DZ, 027F4ZZ, 027G04Z, 027G0DZ, 027G0ZZ, 027G34Z, 027G3DZ, 027G3ZZ, 027G44Z, 027G4DZ, 027G4ZZ,

027H04Z, 027H0DZ, 027H0ZZ, 027H34Z, 027H3DZ, 027H3ZZ, 027H44Z, 027H4DZ, 027H4ZZ, 027J04Z, 027J0DZ, 027J0ZZ, 027J34Z, 027J3DZ, 027J3ZZ, 027J44Z, 027J4DZ, 027J4ZZ, 02RF07Z, 02RF08N, 02RF08Z, 02RF0JZ, 02RF0KZ, 02RF37H, 02RF37Z, 02RF38H, 02RF38N, 02RF38Z, 02RF3JH, 02RF3JZ, 02RF3KH, 02RF3KZ, 02RF47Z, 02RF48N, 02RF48Z, 02RF4JZ, 02RF4KZ, 02RG07Z, 02RG08N, 02RG08Z, 02RG0JZ, 02RG0KZ, 02RG37H, 02RG37Z, 02RG38H, 02RG38N, 02RG38Z, 02RG3JH, 02RG3JZ, 02RG3KH, 02RG3KZ, 02RG47Z, 02RG48N, 02RG48Z, 02RG4JZ, 02RG4KZ, 02RH07Z, 02RH08N, 02RH08Z, 02RH0JZ, 02RH0KZ, 02RH37H, 02RH37Z, 02RH38H, 02RH38N, 02RH38Z, 02RH3JH, 02RH3JZ, 02RH3KH, 02RH3KZ, 02RH47Z, 02RH48N, 02RH48Z, 02RH4JZ, 02RH4KZ, 02RJ07Z, 02RJ08N, 02RJ08Z, 02RJ0JZ, 02RJ0KZ, 02RJ37H, 02RJ37Z, 02RJ38H, 02RJ38N, 02RJ38Z, 02RJ3JH, 02RJ3JZ, 02RJ3KH, 02RJ3KZ, 02RJ47Z, 02RJ48N, 02RJ48Z, 02RJ4JZ, 02RJ4KZ

■ ICD9:

- 3940, 3941, 3942, 3949, 3950, 3951, 3952, 3959, 3960, 3961, 3962, 3963, 3968, 3969, 3970, 3971, 3979, 4240, 4241, 4242, 4243

■ DRG:

- 216, 217, 218, 219, 220, 221, 266, 267, 319, 320

■ CPT:

- 0345T, 0483T, 33361, 33362, 33363, 33364, 33365, 33366, 33367, 33368, 33369, 33390, 33391, 33405, 33406, 33410, 33411, 33412, 33413, 33418, 33419, 33420, 33422, 33425, 33426, 33427, 33430, 33440, 33460, 33463, 33464, 33465, 33468, 33470, 33471, 33474, 33475, 33477

■ SNOMED

- 368009, 11851006, 195005009, 409712001, 79619009, 86466006, 838448003, 48724000, 194733006, 270906004, 373116009, 838450006, 194736003, 31085000, 40023000, 838449006, 266250003, 233860003, 194867006, 286031000119106, 838451005, 838453008, 194726006, 703359009, 1155719000, 194978002, 708121009, 194727002, 838452003, 787001, 371813006, 703259006, 371814000, 371816003, 371815004, 1142308000, 703320000, 703330009, 703328007, 8722008, 60573004, 60234000, 836480008, 836482000, 838544003, 836481007, 12023003, 427515002, 838546001, 838545002, 194987006, 703323003, 703325005, 703236000, 233863001, 195002007, 195005009, 195003002, 195004008, 194734000, 194735004, 16440002, 370141003, 59464004, 71799002, 81552002, 20721001, 111287006, 67696008, 838454002, 838456000, 703319006, 194990000, 838455001, 194741006, 49915006, 838535004, 838537007, 838536003, 194991001, 88318005, 194992008, 76267008, 56786000, 251006007, 251007003, 91434003

b) Has any sensitivity analysis been conducted to confirm the reliability of self-reported diagnoses? A study reports that ~10% of AHMS participants have shared clinical health records (<https://doi.org/10.1038/s41746-024-01187-5>). If this holds true, the vast majority of disease diagnoses rely on self-reports from follow-up surveys, and the sensitivity of self-reported incident diseases is shown to be substantially low, raising questions about the validity and accuracy of reporting. It may also be influenced by factors such as age or education level.

In supplemental section S4 we break down the source of each event label, including counts of how many are derived from self report vs health records. For instance, of the new diabetes diagnoses, 658 are derived from monthly surveys, and 370 are derived from clinical health records. Among the survey responders that positive identify a new diagnosis of diabetes, about 19% (127 of 658) also have an elevated a1c in their clinical health records. We add this discussion into Section S4, along with the counts and data definitions.

In general, we note that we are more likely to miss positive cases than to introduce spurious ones into our analysis. Because of the positive association between PpgAge and increased risk of these adverse events, these noisy labels likely result in an

under-estimation of the strength of this association.

We do want to emphasize that our use of self reported medical history and clinical events is certainly a limitation of this study. Additionally, the naturalistic nature of this data is both an asset (i.e., we collect very noisy, real world data), but an additional limitation when interpreting the strength of these associations. This is observational data, but our analysis — which adjusts for a variety of relevant factors — suggests that there is a strong association between PpgAge gap and incident disease that warrants further investigation.

c) Please consider medication use in the identification of incident disease.

Clinical outcome labels are currently limited to self-reported surveys, labs, diagnosis codes, and claims. When developing the coding phenotypes for the three incident cardiac diseases, we consulted with a medical coding expert and determined that there were no new medications that are specific enough to rule in a new cardiac diagnosis. It's certainly true that new medications could furnish new diabetes events (e.g., those without diagnosis codes or labs, but with new DM medications). As of now, we focus on just using labs and medical coding data for simplicity. Similar to the description above, introducing potentially spurious controls would likely weaken the association between PpgAge gap and event occurrence. This is a good point, and we will work toward using and analyzing medications in future research.

3. In Fig 4, the Cox PH models use a dichotomised group of >6 years vs. ≤6 years for PpgAge gap. It is unclear for choosing this 6-year cutoff in this analysis, especially since a continuous age gap may be more relevant for application and provide interpretable insights.

The Cox model is dichotomized only for reporting purposes. The model was fit using a continuous-valued PpgAge gap covariate. Instead of reporting the effect of e.g., 1 standard deviation, we reported the effect of >6 years to be consistent with the relative risk analysis cutoffs used. We now clarify this in the survival analysis paragraph in the Methods section.

4. In Figure S1, PpgAge gap is relatively symmetric around 0 years, and the BA plots show 11.96SDI of ~6 years in the healthy test set and ~8 years in the general population. This suggests that the individuals with PpgAge gap > 6 years are likely outliers, very fast-aging looking comparing to the rest of the population. Please clarify.

The PpgAge >6 year cutoff constituted 11.6% (14,027 of 120,235) of participants in the general cohort. In this same cohort the prevalence of

- Elevated blood pressure diagnosis is 22.7% (27,286 participants)
- High cholesterol diagnosis is 24.7% (29,747 participants)
- Diabetes diagnosis 6.1% (7,319 participants)
- Heart disease diagnosis is 2.3% (2,819 participants)
- Heart attack is 1.5% (1,832 participants)
- Heart failure diagnosis is .83 % (999 participants)

By comparison, the PpgAge >6 year prevalence is smaller than the prevalence of hypertension and hyperlipidemia, but nearly twice as large as the prevalence of diabetes, and 5 to 10 times as large as those with heart disease related diagnoses. This cutoff certainly represents an elevated stratum (a little more than the top 10%) but is certainly not exclusively outliers. We include these statistics in the description of the survival analysis in the Results section.

5. The section reporting incident cardiometabolic disease is relatively brief. More detailed results and discussion would be appreciated.

We have added more discussion of the incident cardiometabolic disease analysis in the discussion and results sections. Additional analysis details have been included in the supplement in Section S4.

6. Various sensitivity analyses could strengthen the results, e.g., stratified analysis by age groups, repeating association testing considering only EHR data, testing for reverse causality.

We now include a sensitivity check by running the survival analysis for each of the four outcomes, stratified into two age subgroups (young vs old). We use the median age of event subjects for each outcome to do this stratification, so that we retain as many events as possible in each subgroup. While we do see a small lessening of the strength of the effect in a couple of the older subgroups, in general we still see a very consistent positive hazard ratio for each PpgAge gap effect, suggesting that the strength of PpgAge gap's effect remains strong across different ages. The results of this sensitivity analysis can be found in Figure S5 in the supplement, which shows the table of hazard ratios and confidence intervals for all 4 outcomes and young/old age cohorts.

7. For KM plots in Fig 4,
a) Please show the number at risk over time by strata.
b) Instead of survival probability, it may be more informative to present cumulative incidence risk.
c) Please include 95% confidence intervals for all curves.
d) In line 227, authors state that "Survival curves stratified by age and PpgAge gap nearly overlap, as well — the survival for a chronological 55 year old with a PpgAge gap of +6 is similar to a chronological 65 year old with a PpgAge gap of (slightly less than) -6." However, Fig 4b is an unadjusted KM curve. Does this hold even after adjusting for covariates? Please include adjusted risk curves.

The original Figure 4 plots depict fitted survival curves from the Cox model, so each curve is defined by a particular set of covariates rather than explicit subgroups or strata as in a KM analysis. To remedy this, we now include a KM analysis, stratified by PpgAge gap subgroup. These plots, available in the supplement in Figure S6, now depict the number at risk over time for each strata (as well as the total number of censored and event subjects), the survival probability, as well as 95% confidence intervals for the survival curves. Throughout, we see strong separation across the PpgAge gap strata.

For the fitted survival curves from the Cox models, when we compare across chronological ages (e.g., 55 vs 65), we use average covariate values across all subjects within 5 years of the presented age (e.g. age 55 uses a 50-60 stratum), and then depict the effect due to changing the PpgAge gap value (e.g., from 0 to 6). In this sense, we are showing a comparison that adjusts for covariates. And with the additional KM analysis described in the previous paragraph, we also now depict unadjusted (marginal) survival curves.

Instead of survival probability, it may be more informative to present cumulative incidence risk.

In our setting, since we are not modeling competing or recurrent events, cumulative incidence is mathematically equivalent to the inverse of the survival probability, i.e., $CIF(t) = 1 - S(t)$. This has been clarified in the caption of Figure 4.

Reviewer #3

The authors satisfactorily addressed all the comments raised by implementing the paper or by providing a convincing response in the rebuttal letter. The manuscript is improved after the round of revision and I recommend to accept the paper.

Thank you for reviewing our revision. The authors appreciate your thorough review and the incorporation of your suggestions has strengthened the manuscript.

Reviewer #4 (Remarks to the Author)

Thank you for addressing my comments in detail - I learnt much from your response. In this report I only focus on one comment which I think may warrant further revision.

Original Comment 1: There is little analysis of how PpgAge changes within an individual over time (i.e. whether it increases by an average of 1 year per chronological year), and I thought this could have greatly enhanced the study. Furthermore, I envisage this would be feasible given that some participants contributed data for four years.

Thank you for including the analysis of PpgAge changes over time (Supplemental Figure S1). In your rebuttal you state that "these age deltas are better calibrated for younger participants". I would suggest going further than this by: (i) including discussion of this in the manuscript (I didn't see any discussion of Figure S1 (d), although perhaps I have missed it); and (ii) commenting on what this tells us about this plot tells us about the potential utility of PpgAge. Personally, I think that if PpgAge rate is not significantly higher than zero for particular age groups (e.g. age 45+ in females, and 55+ in males), then it raises questions about the ability of PpgAge to track changes in healthy ageing in these groups. In addition, if PpgAge rate differs significantly between age groups, then perhaps this tells us that the potential applications of PpgAge may differ between age groups (e.g. tracking healthy ageing in younger age groups, vs. stratifying according to disease risk in older age groups). I think some discussion of this would be helpful.

Thank you for digging into this part of the analysis.

The calibration of any aging clock across age groups is a pervasive issue. Following [Tian et al 2023](#) and [Smith et al 2019](#), we do a post hoc adjustment to PpgAge deltas to reduce the regression to the mean effect, which simplifies comparisons of PpgAge gap across different chronological age groups. All age clocks based on predicting chronological age (or related surrogates) will suffer from this same problem, which is rooted in the predictor's regression to the mean. That is, older individuals tend to be under predicted, whereas younger individuals tend to be over predicted. This regression to the mean phenomenon frustrates the interpretation of predicted age in terms of years across chronological age groups, hence the frequently used post hoc adjustment described in [Smith et al 2019](#), which we describe in the Methods section.

We similarly expect that the calibration of age *rate* is sensitive to chronological age group. It is not obvious how to adjust for this quantity, however, and would likely require a more sophisticated (and potentially novel) statistical model or procedure to directly describe variability of observations within subject. We also note that the plots in Figure S1 reveal *average* age rate within each age group. This hides a lot of the variability — some individuals do indeed “age” at rates closer to 1, whereas some are negative. Because the average of this uncalibrated statistic is close to zero does not indicate that the uncalibrated statistic is not useful. Our goal is to address the temporal calibration issue and study its impact in a future publication.

The potential utility of PpgAge is shown in the cross sectional and survival analyses — e.g., PpgAge at baseline is highly associated with a range of disease and remains predictive of future adverse cardiovascular events. This is shown across all age groups (though the strength of the effect is, of course, heterogeneous across age groups — see **Figure 4**).

We discuss Figure S1 and include the above points in the Discussion section. We also note that you have highlighted some exciting new directions for research that we look forward to pursuing.

Response To Reviewers (NCOMMS-24-37590C)

REVIEWER #1 (REMARKS TO THE AUTHOR):

Thank you for thoroughly addressing the comments and improving the manuscript's clarity. The authors have sufficiently evaluated and responded to my comments.

Thank you again for your feedback.

Reviewer #1 (Remarks on code availability):

In the code provided, 'analysis/generate_synthetic_data.py' mentioned in README file is missing.

Apologies, we are re-submitting the full repository with this submission with an updated README to more clearly describe the way synthetic data are generated and the example pipeline can be run.

REVIEWER #4 (REMARKS TO THE AUTHOR):

Thank you for addressing my comments.

Thank you again for all of your thoughtful feedback.s

Reviewer #4 (Remarks on code availability):

My apologies, I don't know why I haven't looked at the code until now.

Please note that whilst I have had a brief look at the code, I have not reviewed it in-depth, partly because I'm not an expert at Python (making it difficult to provide a well-informed critique), and partly because I'm not an expert in all the packages and statistics used. I have also only reviewed the python code, and I have not reviewed the R-code (I've barely ever used R).

I note that as far as I can tell the code does not allow one to derive PpgAge from a PPG pulse wave - rather it allows one to perform the statistical analysis after PpgAge has been derived. Furthermore, it doesn't not allow one to recreate the original results, rather it uses synthetic data.

Yes — we originally included only the software to reproduce our analysis. We have now added a PPG representation learning script that initializes a model and trains it with the contrastive loss + augmentations on synthetic (or user provided) data. We are currently unable to share actual real biosignal and study participant data, but hope that these scripts enable one to reproduce a similar analysis on a new dataset.

I have the following comments:

- The code appeared well structured and accompanied by a helpful readme file.*
- I was able to get the code to run up until a point (I encountered an error in analysis_3_pregnancy.py)*
- Might it be helpful to include a requirements.txt file? I didn't immediately see a list of required packages.*
- The readme states that you can generate synthetic dataframes using 'analysis/generate_synthetic_data.py'. I didn't find this python file. However, I presume that I did generate synthetic dataframes, as I was able to create some csvs and plots.*
- Generally, it is helpful to be able to follow the statistical analysis, even if only using synthetic data. However, it would be even more helpful if one was able to reproduce the original analysis, including not only the statistical analysis, but also the derivation of PpgAge from PPG pulse waves. Of course, I acknowledge that there are potential difficulties in facilitating this, e.g. (i) providing code for derivation of PpgAge may not be possible because of IP restrictions; and (ii) providing the original*

data is subject to the requestor going through the (lengthy) process outlined in the 'Data Availability' statement. In my view, this is simply a limitation of the ability of reviewers to critique the work (a limitation which is often, although not always, encountered in similar research). I would be happy for the manuscript to be accepted despite this limitation.

We have updated the README with instructions on how to run our pipeline, and fixed all errors encountered. We include a requirements.txt package, and instructions on creating a custom environment. One can now run the steps of the pipeline in order: (i) produce ppg representations from raw (synthetic) ppg data, (ii) produce an age model and age predictions, (iii) analyze predictions of the age model.

We appreciate your understanding, and acknowledge that we are currently unable to share participant data to make the work truly turnkey reproducible. We hope that our expanded software pipeline will enable a softer form of reproducibility that is still useful to both the reviewers and readers.

Response to Reviewers

NCOMMS-24-37590E

Reviewer #4 (Remarks to the Author):

Thank you very much for tidying up the code. The README file reads well.

Thank you again for reviewing the manuscript and the code.

I have tried following the code, and wasn't able to. My apologies - I suspect this demonstrates my lack of familiarity with Python, rather than issues with the code. Therefore, I would suggest the following to the Editors:

- If you want the code reviewed, then I would suggest that someone with a good working knowledge of Python could check to see whether the code can be run. This could potentially be someone without knowledge of the research topic. I suspect that the errors I've encountered might be overcome by someone skilled in Python programming. If you want my input then I will be happy to provide input where possible (acknowledging that I am not an expert in Python).*
- You may also decide that the code review has gone sufficiently far, and that you can make a decision without requiring further review.*

In case it's helpful, the issues I encountered were:

- I received errors when running 'pip install -r requirements.txt'. e.g. 'ERROR: Could not find a version that satisfies the requirement torch==2.4.1'. I tried installing each package manually (e.g. 'pip install torch'.*
- I then received an error when running 'python -m training.main -c TrainingPpgSsl', e.g.*

*... torch/utils/data/_utils/signal_handling.py", line 66, in handler
_error_if_any_worker_fails()*

RuntimeError: DataLoader worker (pid 19944) is killed by signal: Terminated: 15.

I suspect that someone skilled in the art of Python might be able to handle these errors.

Unfortunately, I don't have the level of understanding to know whether these can be handled easily or not.

Thank you for being so hands on with our code. And apologies for the confusion. Below, and in the repository, I have replicated detailed steps for replicating installation and running the example scripts. I have also provided a 'pip freeze' example to show exact versions of all packages of my system. This information is also now available in the repository, and reproduced below.

```
%> conda --version  
conda 24.11.3
```

```
(ppg-age) %> python --version  
Python 3.11.0
```

```
(ppg-age) %> pip freeze  
asttokens==3.0.0  
autograd==1.8.0  
autograd-gamma==0.5.0  
contourpy==1.3.3
```

cycler==0.12.1
decorator==5.2.1
executing==2.2.0
filelock==3.18.0
fonttools==4.59.0
formulaic==1.2.0
fsspec==2025.7.0
interface-meta==1.3.0
ipython==9.4.0
ipython_pygments_lexers==1.1.1
jedi==0.19.2
Jinja2==3.1.6
joblib==1.5.1
kiwisolver==1.4.8
lifelines==0.30.0
lightning-utilities==0.15.2
MarkupSafe==3.0.2
matplotlib==3.10.5
matplotlib-inline==0.1.7
mpmath==1.3.0
narwhals==2.0.1
networkx==3.5
numpy==2.3.2
packaging==25.0
pandas==2.3.1
parso==0.8.4
patsy==1.0.1
pexpect==4.9.0
pillow==11.3.0
prompt_toolkit==3.0.51
ptyprocess==0.7.0
pure_eval==0.2.3
Pygments==2.19.2
pyparsing==3.2.3
python-dateutil==2.9.0.post0
pytz==2025.2
scikit-learn==1.7.1
scipy==1.16.1
seaborn==0.13.2
six==1.17.0
stack-data==0.6.3
statsmodels==0.14.5
sympy==1.14.0
threadpoolctl==3.6.0
torch==2.8.0
torchaudio==2.8.0
torchmetrics==1.8.1
torchvision==0.23.0
traitlets==5.14.3
typing_extensions==4.14.1
tzdata==2025.2
wcwidth==0.2.13
wrapt==1.17.2